L02 — study notes

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Lecture 2 — Static games II: Nash equilibrium and mixed strategies

· Map of the notes

Taught unit. Built on 2026-10-16 from the study guide; reconciled on 2026-10-25 with `Aula/Lecture02.pdf` (slides 1–56). Notation follows the slides: payoffs u_i (Tadelis writes v_i), mixed strategies $\sigma_i \in \Sigma_i = \Delta(S_i)$, Cournot benchmarks q^m and q^{cp}, monopoly price p^m. Sections the lecture did not teach are kept as reading and flagged **"Not covered in class"**; the reconcile table is below.

One claim. Lecture 1 asked what a rational player would never do; Lecture 2 asks what he might do: a best response to some belief, and, iterated, the rationalizable strategies. **Nash equilibrium adds one demand, that the beliefs be correct**: every player best-responds to what the others actually play, which makes the profile self-enforcing and a fixed point of the best-response correspondences. **With continuous strategies the equilibrium is the crossing of best-response functions**: Cournot lands between monopoly and competition ($q_i^* = \frac{2}{3}q^m$), Bertrand on marginal cost (the Bertrand paradox). **Players can also mix, and by the principle of indifference each player's mix is pinned by the other player's indifference** (the Battle of the Sexes gains a third, mixed equilibrium). **Nash's theorem closes the loop: every finite game has an equilibrium, possibly mixed**, because mixed strategies make the fixed-point argument work.

Three layers in this unit:

Layer	What it means here
U — intuition	Gibbons's motivation (self-enforcing predictions, conventions), Tadelis's stories (Battle of the Sexes, Stag Hunt, Hardin's commons, Hotelling's parties), mixing as uncertainty rather than coin-flipping
C — course (examinable)	what the slides teach: best response, BR correspondence, never a best response, rationalizability (Tadelis Defs. 4.5–4.8, Prop. 4.4); NE (Def. 5.1) and the slides' Props. 1–2 with proofs; Cournot (duopoly, cartel, n firms) and Bertrand (BR correspondence, unique (c, c)); mixed strategies, expected payoffs, mixed NE, the **principle of indifference** (an iff, proved on slides 45–46; = MWG Prop. 8.D.1); the Battle of the Sexes in mixed strategies; Nash's theorem (statement)

Layer	What it means here
G — formal / reading	Tadelis material the lecture skipped (the commons, Hotelling voting, mixed dominance and Prop. 6.2, the Kakutani proof, the cent-grid and asymmetric Bertrand) and MWG §8.C–8.D, App. A. **Not examinable** unless a problem set brings it

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Reconcile with the slides (2026-10-25)

Slides	What they teach	Where in the notes	Change
2–8	recap: DSE, Pareto, epistemic IESDS	Lecture 1 notes	linked only
9–12	BoS; best response, BR correspondence, NBR; rationalizability; Prop. (= Tadelis 4.4); Lessons 3–4	01 §1–§2, §5	aligned; Lessons 3–4 added
14–18	NE; interpretations; Props. 1–2 with proofs; weak-dominance caveat; road game; PD	02	**added** §2b–§2d
20–27	Cournot in q^m, q^{cp}; BR; NE $\frac{2}{3}q^m$; IESDS sequence; cartel matrix; n firms	03	**added** §0, §4b, §4c
29–40	Bertrand with a, b, c; BR correspondence; (c, c) unique in two steps; paradox; Cournot or Bertrand	04	**added** §0; cent grid, unequal costs and differentiated products marked not covered
42–51	mixed strategies; expected payoffs; mixed NE; principle of indifference (iff, proof); BoS mixed; interpretations; Nash's theorem	07, 09 §1	**added** §2b, §2c, §6b, §7; MWG 8.D.1 promoted from G to C
53–55	mini case: monitoring remote work (graded, in pairs)	07 §9	**added**, method only (no solution)

Slides	What they teach	Where in the notes	Change
—	not on the slides: beauty contest, Stag Hunt, commons, Hotelling voting, Matching Pennies, RPS, §6.2.3 game, mixed IESDS, Kakutani proof	01 §4, 02 §4, 05, 06, 07 §1/§3–§6, 08, 09 §2–§6	kept, flagged **Not covered in class**

Reading order

Part I — beliefs and best responses (Tadelis §4.3)

1. L02-01-best-response-rationalizability — slides 9–12; §4.3, pp. 69–76: Defs. 4.5–4.8, Props. 4.3–4.4 proved, Lessons 3–4; the best-response **correspondence** (ties, p. 72); **rationalizability in Cournot converging to 30**; why it says "anything can happen" in the Battle of the Sexes. (Beauty contest: not covered.)

Part II — Nash equilibrium in pure strategies (Tadelis Ch. 5)

2. L02-02-nash-equilibrium — slides 14–18; §5.1–5.2.1, pp. 79–84: Def. 5.1, the **two requirements**, the slides' **Props. 1–2** (NE survives IESDS; a unique IESDS survivor is the NE) proved; NE with weakly dominated strategies; the road game and the PD; the **underline method**; multiple equilibria and focal points.
3. L02-03-cournot — slides 20–27; §5.2.3, pp. 87–88: $q_i^* = q^m - \frac{1}{2}q_j$, NE $\frac{2}{3}q^m$ (30 in the example); $\mathbf{45 < 60 < 90}$; the **cartel matrix** (a prisoner's dilemma); n **firms**, $Q^* \to q^{cp}$.
4. L02-04-bertrand — slides 29–40; §5.2.4, pp. 88–93: the best-response **correspondence** with $p^m = (a+c)/2$; (c,c) **is the unique NE** (two steps); the **Bertrand paradox**; Cournot or Bertrand? (Cent grid, unequal costs, differentiated products: not covered in class.)
5. L02-05-commons — **not covered in class**. §5.2.2, pp. 84–87: $K/3$ vs $K/4$; **both gain** $\ln(9/8)$; Gibbons's goats and the proof that $G^* > G^{**}$.
6. L02-06-hotelling-voting — **not covered in class** (slide 3 only previews the pharmacies with prices). §5.2.5, pp. 93–95: interval best responses; **both candidates at the median**; the link to Lecture 1's pharmacies.

Part III — mixed strategies and existence (Tadelis Ch. 6)

7. L02-07-mixed-strategies — slides 42–55; §6.1–6.2, pp. 101–114: Defs. 6.1–6.6, the **principle of indifference** (iff, both directions proved); the **Battle of the Sexes,** $(p,q) = (\frac{2}{3}, \frac{1}{3})$, each expecting $\frac{2}{3}$; interpretations; the **mini case** method. (Matching Pennies, RPS, the §6.2.3 game: reading.)
8. L02-08-iesds-rationalizability-revisited — **not covered in class**. §6.3, pp. 114–117: **dominance by a mixture**, best responses to mixed beliefs, the 3×3 game reduced in two rounds; **Prop. 6.2**.
9. L02-09-existence — slide 51 states **Nash's theorem**; the proof is **not covered in class**. §6.4, pp. 117–123: **NE = fixed point of** BR; Brouwer, Kakutani's four conditions checked for 2×2 games.

Support

10. L02-10-formulas — the study-guide content mapped to notes; every definition with its number; every number.
11. L02-11-pitfalls — fifteen confusions with counterexamples; **30 original True/False items** (25–30 on the slides' material).
12. L02-12-python — best responses, rationalizability, Cournot, cent-grid Bertrand, voting, support enumeration, the dominance LP and the fixed-point check, all asserted against the books.

The one result to hold

σ^* is a Nash equilibrium $\iff$ $\sigma_i^* \in BR_i(\sigma_{-i}^*)$ $\forall i$ $\iff$ $\sigma^* \in BR(\sigma^*)$, and every s_i in the support of σ_i^* earns $u_i($

Best response to correct beliefs; a fixed point; indifference on the support. The first clause defines the concept, the second proves it exists, the third computes it.

Numbers and cases to have memorized

Case	Result	Note
Cournot $P = 100 - Q$, $c = 10$ (slides 20–24)	$q_i^* = 30 = \frac{2}{3}q^m$, $Q^* = 60$, $P^* = 40$, $\Pi^* = 900$; $q^m = 45$; $q^{cp} = 90$	03
Cournot cartel (slide 25)	22.5 each, 1,012 each; deviation $q_i^*(22.5) = 33.75$ earns 1,139	03
Cournot, n firms (slide 27)	$q_i^* = \dfrac{2q^m}{n+1}$, $Q^* = \dfrac{2n}{n+1}q^m \to q^{cp}$	03
Bertrand (slides 29–38)	$p^m = \frac{a+c}{2}$; unique NE (c, c), profits 0	04
Battle of the Sexes (slides 47–50)	(O, O), (F, F), and $(p, q) = (\frac{2}{3}, \frac{1}{3})$; payoff $\frac{2}{3}$ each	07
Road game, PD (slide 18)	(L, L), (R, R); PD unique NE (D, D)	02
Reading: commons; Hotelling voting	NE $K/3$, optimum $K/4$; both at the median	05, 06
Reading: Matching Pennies; RPS; §6.2.3 game	$(\frac{1}{2}, \frac{1}{2})$; $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$; (M, R), (D, C), $p = \frac{1}{6}$, $q = \frac{3}{7}$	07

Figures

Built by `figures/make_figures.py` (run: `python Notas/L02/figures/make_figures.py`; it ends with asserts on the plotted numbers).

Figure	Shows	Lives in
L02-underline-method	the p. 81 game with both players' best replies marked; (M, C) the only doubly marked cell	L02-02-nash-equilibrium §3
L02-cournot-best-responses	Fig. 5.2: BR lines, NE (30, 30), rationalizability squares, the collusive split and its deviation	L02-03-cournot §3, L02-01-best-response-rationalizability §3
L02-bertrand	cent-grid best response with the two NE; Fig. 5.3 with $c_1 = 1$, $p_2 = 2$	L02-04-bertrand §2
L02-commons	Fig. 5.1 and the symmetric payoff peaking at $K/4$	L02-05-commons §2
L02-hotelling-voting	the interval best responses on 101 platforms, unique NE at 0	L02-06-hotelling-voting §2
L02-bos-mixed	slides 47–50: Alex's payoffs $2q$ and $1-q$ crossing at $q = \frac{1}{3}$; both best responses crossing three times, mixed at $(p, q) = (\frac{2}{3}, \frac{1}{3})$	L02-07-mixed-strategies §2c
L02-mixed-2x2	the §6.2.3 game: indifference at $q = \frac{3}{7}$; Fig. 6.4 with three crossings	L02-07-mixed-strategies §5
L02-mixed-dominance	Gibbons Figs. 1.3.1–1.3.2: dominated by a mix; a best reply only to mixed beliefs	L02-08-iesds-rationalizability-revisited §2

Interactive companions

- The Mixing Square (`companions/companion-mixed-2x2.html`) — any 2×2 game through the four payoff differences that decide every best response $(\alpha_1, \beta_1, \alpha_2, \beta_2)$, plus two belief probes q, p; draws both best-response correspondences in the (q, p) square and the two indifference lines, lists **every NE (pure, mixed, or a continuum) exactly**, and reads out $\hat{q}$, $\hat{p}$, each player's gain at the probe and Gibbons's case. Default: Tadelis §6.2.3; presets Matching Pennies, Battle of the Sexes, Stag Hunt, prisoner's dilemma. Linked from notes 02 and 07.
- The Cournot Crossing (`companions/companion-cournot-best-responses.html`) — a, b, c_1, c_2 move the two best-response lines; reads out (q_1^*, q_2^*), price, profits, the monopoly and competitive benchmarks, the exit threshold and the collusion deviation. Default: Tadelis p. 87. Linked from notes 01 and 03.

Both are checked by `companions/check_companions.py` (Node runs each page's model; Python recomputes independently).

Scope and ceiling

- **Examinable (taught):** what the slides cover, at the C layer: the definitions (Tadelis's numbering), the slides' Props. 1–2 and the principle of indifference **with their proofs**, Nash's theorem as a statement, and solving matrix games (pure and mixed), Cournot (duopoly, cartel, n firms) and Bertrand.

- **Reading, not covered in class:** the commons, Hotelling voting, the beauty contest, the Stag Hunt, Matching Pennies / RPS / the §6.2.3 game, mixed dominance and Prop. 6.2, the fixed-point proof, the cent-grid, asymmetric and differentiated Bertrand. Useful practice; examinable only if a problem set brings them.
- **Named only:** the index theorem (odd number of equilibria, Tadelis footnote 6); Harsanyi's purification (returns in Lecture 7); Gibbons §1.2.C final-offer arbitration (pp. 22–26), a NE in a game against an uncertain arbitrator, not written up here; the MWG G layer.
- **Out of scope** (CLAUDE.md): mechanism design, cheap talk, moral hazard. Correlated equilibrium and refinements (trembling-hand, MWG §8.F) are not part of this lecture.
- Exercises are cited by number and page only.

Course-wide position

Lecture	Relation to this one
Lecture 1	dominance and IESDS: what Lecture 2 sharpens (NE $\subseteq$ IESDS survivors) and extends (mixed dominance); the pharmacy case is Hotelling
Lecture 2	**best responses, rationalizability, Nash equilibrium, mixed strategies, existence**
Lecture 3	dynamic games: NE admits non-credible threats; subgame perfection refines it; Stackelberg vs this Cournot
Lecture 4	multistage games and bargaining: NE in every subgame
Lecture 5	repeated games: can repetition sustain the Pareto outcome a one-shot NE misses (commons, PD)?
Lecture 6	collusion in Cournot and Bertrand: the deviation of L02-03-cournot §4 against future punishment
Lecture 7	Bayesian NE: best responses to beliefs about types; purification of mixed strategies
Lecture 8	signaling and PBE: NE plus beliefs that must be consistent

Rules pack: rules/game-theory.

Bibliography actually used

Book	Where
Tadelis (2013) §4.3, Ch. 5, Ch. 6 (pp. 69–126)	the spine of every note; numbering of every definition
Gibbons (1992) §1.1.C (motivation, App. 1.1.C), §1.2.A–B, §1.2.D, §1.3.A–B	U layer; Cournot collusion; differentiated Bertrand; the goats; mixed dominance figures; the 2×2 cases
Cabral (2017) §7.1 (pp. 161–167)	best responses marked by asterisks; focal points; footnote f on mixed beliefs

Book	Where
MWG (1995) §8.C–8.D, App. A (pp. 242–261)	the G layer

Related project material. Lecture 2: `Listas/pratica-lecture-02.md` (book exercises, tiered), `Simulados/quiz-lecture02-{1,2,3}-*.md`, `NotebookLM/audio/lecture-02-audio-part{1,2}.md`, `Leituras/lecture-02-notes-narrated.txt`. Lecture 1's material is the prerequisite: `Listas/pratica-lecture-01.md`, `Listas/lista-lecture-01.tex` (with `Listas/lista-lecture-01-check.py`), `Simulados/quiz-lecture01-{1,2,3}-*.md`. After class, the study guide's drill: Tadelis 5.2, 5.3, 5.15 (pp. 95, 99), 6.2, 6.3 (p. 123); Gibbons 1.3, 1.5, 1.8 (pp. 49–50); MWG 8.C.1, 8.D.3–8.D.5 (p. 263); Muñoz-Garcia & Toro-Gonzalez Ch. 2 (pp. 37–84), Ch. 3 (pp. 85–151).

Slides 9–12 · 4.3 · Best responses, beliefs and rationalizability: asking what a rational player *might* do

> **Claim.** Lecture 1 removed what a rational player would **never** do (dominance, IESDS). Rationalizability asks the mirror question: what **might** a rational player do? The answer is a best response **to some belief**. Iterating "only strategies that are a best response to some belief about strategies that are themselves rationalizable" under common knowledge of rationality is the **same machine as IESDS in two-player games** (L02-08-iesds-rationalizability-revisited). It has bite in Cournot (it converges to 30) and in the beauty contest (it ends at $\{0, 1\}$), and none at all in the Battle of the Sexes.

Prev: Lecture 1 map · Next: L02-02-nash-equilibrium · Map: L02-00-map

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1 · U — why beliefs enter

In a decision problem the best action depends only on the problem. In a game it depends on **what the others do**, and in a static game a player cannot see that, so the player must hold a **belief** (Tadelis p. 71). Tadelis's test of whether someone is irrational is to ask *"What were you thinking?"*: if some belief about the others makes the choice optimal, the choice is **rationalized** (p. 69).

The lecture opens this part with Lecture 1's epistemic reading of IESDS (slide 7: S_i^1 is what rationality allows, S_i^k what rationality plus $k-1$ levels of belief in rationality allow, S_i^∞ what common belief in rationality allows) and one question: **"But what if no strategy is dominated?"** Its answer is Example 1, the Battle of the Sexes (slide 9; Tadelis p. 69; Alex picks the row, Chris the column; Alex prefers the opera, Chris football, and being apart is the worst outcome for both):

Alex \ Chris	O	F
O	2, 1	0, 0
F	0, 0	1, 2

Slide 9 asks three questions; the answers, step by step. *Dominant strategy?* Alex's O pays 2 against O but 0 against F, while F pays 0 and 1: neither beats the other against **both** columns, so Alex has none, and by the mirror argument neither has Chris. *Dominated strategies?* None, for the same reason. *What does IESDS predict?* It removes nothing, so all four cells survive: Lecture 1's tools say "anything can happen". Alex's best choice is O if Chris goes to O and F if Chris goes to F; the best choice **flips with the belief**.

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2 · C — the definitions, exactly

Definition 4.5 (p. 70; slide 10). $s_i \in S_i$ is player i's **best response** to $s_{-i} \in S_{-i}$ if

$$u_i(s_i, s_{-i}) \;\geq\; u_i(s_i', s_{-i}) \qquad \forall s_i' \in S_i.$$

Claim 4.2 (p. 70). A rational player who believes the opponents play s_{-i} always chooses a best response to s_{-i}.

Worked (p. 70): if Chris believes Alex goes to the opera, $u_2(O, O) = 1 > 0 = u_2(O, F)$, so Chris's best response is O.

Proposition 4.3 (p. 70). If s_i is strictly dominated, it is not a best response to any s_{-i}.

Proof, every step. Strict dominance gives some s_i' with $u_i(s_i', s_{-i}) > u_i(s_i, s_{-i})$ for **all** s_{-i}. A best response to a particular s_{-i} needs $u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i})$ for that s_{-i}. The two inequalities contradict each other at every s_{-i}, so no such s_{-i} exists. ■

Proposition 4.4 (p. 70; slide 12, stated as "Proposition"). In a finite game, if s^* is a strict dominant-strategy equilibrium, or the unique survivor of IESDS, then $s_i^* \in BR_i(s_{-i}^*)$ for every i.

Proof (Tadelis's contradiction, pp. 70–71). A strict dominant-strategy equilibrium uniquely survives IESDS (Lecture 1), so it suffices to treat the unique survivor. Suppose s_i^* is **not** a best response to s_{-i}^*: some s_i' has $u_i(s_i', s_{-i}^*) > u_i(s_i^*, s_{-i}^*)$. That s_i' was eliminated at some round, while s_{-i}^* was never eliminated, so the strategy s_i'' that eliminated it does even better against s_{-i}^*: $u_i(s_i'', s_{-i}^*) > u_i(s_i', s_{-i}^*) > u_i(s_i^*, s_{-i}^*)$. Repeat; finiteness ends the chain at a strategy that beats s_i^* against s_{-i}^* and **is never eliminated**, contradicting uniqueness of the survivor. ■

Definition 4.6 (p. 72). A **belief** of player i is a profile of opponents' strategies $s_{-i} \in S_{-i}$. (Here a belief is a *pure* profile; Lecture 2's §6 replaces it by a probability distribution, Def. 6.4, p. 105.)

Definition 4.7 (p. 72; slide 10). The **best-response correspondence** selects, for each s_{-i}, the set $BR_i(s_{-i}) \subseteq S_i$ of best responses.

> **It is a correspondence, not a function, because a belief can have several best responses.**

Worked: the 3×3 game of p. 72.

P1 \ P2	L	C	R
U	3, 3	5, 1	6, 2
M	4, 1	8, 4	3, 6
D	4, 0	9, 6	6, 8

Against R, player 1's payoffs are U 6, M 3, D 6, so $BR_1(R) = \{U, D\}$. Against L: U 3, M 4, D 4, so $BR_1(L) = \{M, D\}$. Against C: 5, 8, 9, so $BR_1(C) = \{D\}$.

Definition 4.8 (p. 73; slide 10, "NBR"). s_i is **never a best response** if there is no belief s_{-i} with $s_i \in BR_i(s_{-i})$.

Rationalizability (p. 73; slide 11). "What might a rational player do?" A rational player only chooses strategies that are a best response to some belief. Using common knowledge of rationality, remove every strategy that is never a best response; in the reduced game, remove again; iterate. The survivors are the **rationalizable** strategies.

> **Lesson 3 (slide 12):** a rational player never plays a strategy that is never a best response.
> **Lesson 4 (slide 12; Claim 4.2):** a rational player who believes the opponents play s_{-i} chooses a best response to s_{-i}. (Lessons 1–2 were Lecture 1's: never play a strictly dominated strategy; individually rational choices can be socially undesirable.) Tadelis postpones the formal definition until beliefs can be mixed (§6.3, p. 114, Def. 6.8) because with pure beliefs only the elimination is too weak (L02-08-iesds-rationalizability-revisited §2 shows a strategy that is a best response only to a *mixed* belief).

Concept	Question asked	Assumption
Strict dominance	is there something always better?	rationality
IESDS	what would a rational player never do, iterated?	common knowledge of rationality
Rationalizability	what can a rational player justify by some belief, iterated?	common knowledge of rationality

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3 · Worked — Cournot revisited (§4.3.4, pp. 73–74)

$p(q) = 100 - q$, $c_i(q_i) = 10q_i$. Firm 1 maximizes $u_1 = (100 - q_1 - q_2)q_1 - 10q_1$.

$$\frac{\partial u_1}{\partial q_1} = 100 - 2q_1 - q_2 - 10 = 90 - 2q_1 - q_2 = 0 \implies q_1 = \frac{90 - q_2}{2}.$$

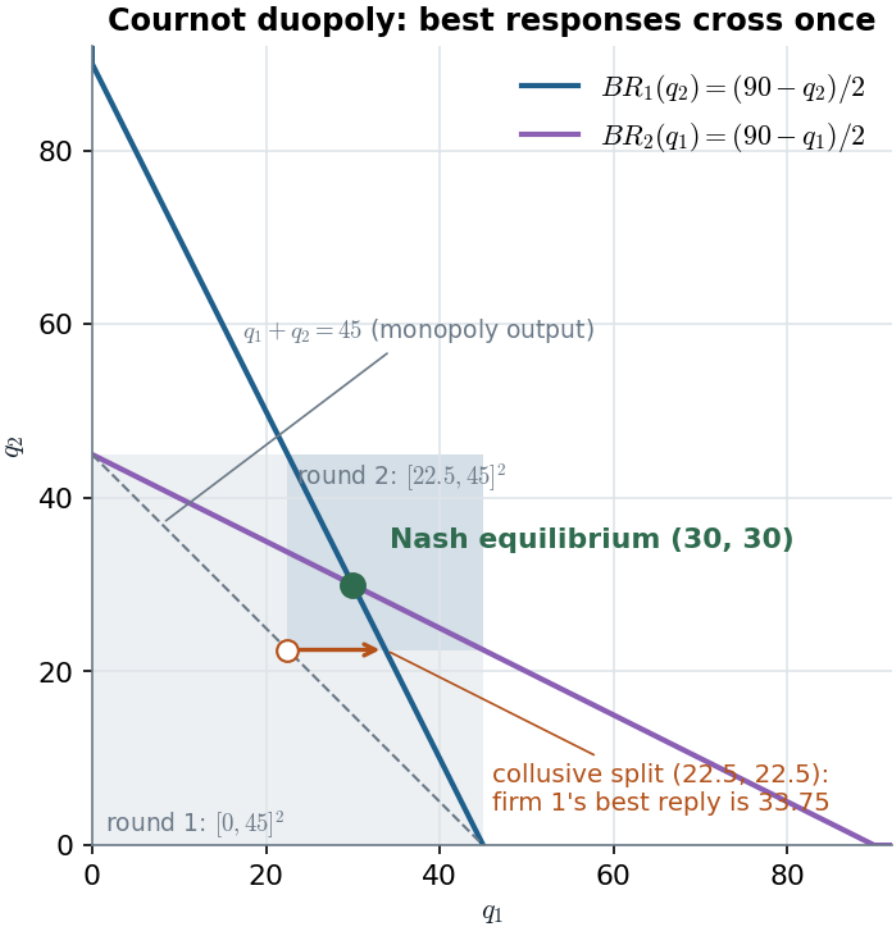
The second derivative is $-2 < 0$, so this is the maximum. Truncated at zero:

$$BR_1(q_2) = \begin{cases} \dfrac{90 - q_2}{2} & 0 \le q_2 < 90 \\ 0 & q_2 \ge 90 \end{cases}$$

(for $q_2 \ge 90$ the price is below 10 at any $q_1 > 0$, so producing loses money). It is a **function**: one best reply per belief.

Round	Beliefs about q_j	Best replies q_i	Why
1	$[0, \infty)$	$[0, 45]$	BR_i is largest, 45, at $q_j = 0$
2	$[0, 45]$	$[22.5, 45]$	$BR_i(45) = 22.5$, $BR_i(0) = 45$
3	$[22.5, 45]$	$[22.5, 33.75]$	$BR_i(45) = 22.5$, $BR_i(22.5) = 33.75$
4	$[22.5, 33.75]$	$[28.125, 33.75]$	$BR_i(33.75) = 28.125$
∞		$\{30\}$	fixed point of $q = (90 - q)/2$: $3q = 90$

Each round halves the width (the slope of BR is $-1/2$), so the intervals shrink to the point $q_i = 30$, which is the Nash equilibrium of L02-03-cournot. Lecture 1 reached the same point by IESDS (Tadelis p. 65; Gibbons pp. 18–20). Gibbons p. 20–21 adds the warning: with **three** firms the process stops after the first round, because Q_{-i} can range over $[0, a - c]$ and every $q_i \in [0, (a - c)/2]$ is the best reply to $Q_{-i} = a - c - 2q_i$.



The shaded squares are rounds 1 and 2 of rationalizability, $[0, 45]^2$ and $[22.5, 45]^2$; they close in on the crossing at (30, 30). Source: Tadelis §4.3.4, pp. 73–74, and Fig. 5.2, p. 88.

Read the figure as a staircase: a belief anywhere in the outer square implies a reply inside the inner one, and so on. The crossing is the only point that survives every round, which is why rationalizability is "unique" here.

> **Interactive companion.** The Cournot Crossing (`companions/companion-cournot-best-responses.html`) — move c_1 and c_2 apart and watch the crossing move along both lines; the rationalizability argument works for any crossing of two lines with slope $-1/2$.

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4 · Not covered in class — the p-beauty contest (§4.3.5, pp. 74–75)

Reading only: the slides do not present the beauty contest; kept as practice in iterated reasoning.

n players choose integers in $\{0, \dots, 20\}$; the winners are those closest to $\frac{3}{4}$ of the average; each player pays 1 and winners split the pot. Tadelis's example: $n = 3$, choices $1, 5, 18$; average 8; $\frac{3}{4} \cdot 8 = 6$; player 2 (at 5) is closest and wins.

Round 1: 20 is never a best response. If you believe the average is below 20, a lower number is closer to $\frac{3}{4}$ of it. If you believe everyone else plays 20, then choosing 20 ties with everyone, while 19 wins alone: the average is $\frac{1}{n}[(n-1)20 + 19] \in [19.5, 20)$, so $\frac{3}{4}$ of it lies in $[14.625, 15)$, and 19 is closer than 20 (footnote 6, p. 75). So 20 is beaten under **every** belief.

Rounds 2–19. In the reduced game the same argument removes the new top number each time, leaving $S_i^{19} = \{0, 1\}$. For $n > 2$ both survive (each is a best response to "everyone else plays it"); for $n = 2$ only 0 survives (p. 75; the proofs are Tadelis exercises 4.8–4.9, p. 78).

> **Common knowledge of rationality is doing all the work.** Tadelis notes that real classes rarely produce a winning number below 4 (footnote 7, p. 75): the prediction is only as good as the assumption that everyone reasons all 19 rounds deep.

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5 · C — evaluating rationalizability (§4.3.6, p. 76)

Criterion	Verdict
Existence	always (as IESDS)
Uniqueness	sometimes sharp (Cournot, slide 24; the beauty contest); in the Battle of the Sexes **"anything can happen"**
Pareto	survivors need not be Pareto optimal (the prisoner's dilemma survives intact)

> **Rationalizability lets each player hold *some* belief; it never asks the beliefs to be right.** In the Battle of the Sexes, Alex can rationally go to O believing Chris goes to O while Chris rationally goes to F believing Alex goes to F. Removing that dissonance is the step to Nash equilibrium (L02-02-nash-equilibrium).

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6 · G — the formal version (MWG §8.C, pp. 242–245)

MWG Def. 8.C.1 (p. 243) defines a best response among **mixed** strategies and "never a best response" against all mixed profiles of the rivals; Def. 8.C.2 (p. 243) calls the survivors of the iterated removal the **rationalizable strategies** (Bernheim 1984, Pearce 1984). The set of rationalizable strategies is no larger than the set surviving IESDS, and the removal order does not matter. Not examinable unless the lecture states it.

Links

- Neighbours: L02-02-nash-equilibrium (the beliefs made correct), L02-08-iesds-rationalizability-revisited (the equivalence with IESDS once beliefs can be mixed).
- Lecture 1 (dominance and IESDS, the same Cournot game by elimination): Lecture 1.
- Rules pack: rules/game-theory items 2, 6.
- Practice (statements in the books): Tadelis 4.6 (p. 77), 4.8, 4.9 and 4.10 (p. 78); Gibbons 1.2 (p. 48); MWG 8.C.1 (p. 263).

Slides 14–18 · 5.1 · Nash equilibrium: best responses to beliefs that are correct

> **Claim.** A Nash equilibrium is a profile in which **every player best-responds to the others' actual strategies**: rationality plus **correct beliefs**. It refines IESDS and rationalizability (every NE survives them; a unique survivor is the unique NE), it is found in a matrix by **marking best replies and looking for a cell marked twice**, it can be multiple (Battle of the Sexes, Stag Hunt), and it need not be Pareto optimal.

Prev: L02-01-best-response-rationalizability · Next: L02-03-cournot · Map: L02-00-map

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1 · U — from rationalizable to self-enforcing

Rationalizability allowed (O, F) in the Battle of the Sexes: Alex at the opera believing Chris goes, Chris at the game believing Alex goes. Each is optimal **given a belief**, but the beliefs are wrong (Tadelis p. 79). Nash's "heroic leap" ties actions and beliefs together: in equilibrium everyone's belief is what the others actually do.

Gibbons's motivation (§1.1.C, pp. 8–9): if game theory offers a **unique** prediction, each player must be willing to play the predicted strategy, i.e. it must be a best reply to the predicted strategies of the others. A prediction that is not a NE is **falsified by the play it predicts**: someone deviates. The same argument applies to a **convention**: a convention that is not a NE is not obeyed. Such a prediction is **self-enforcing**.

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2 · C — the definition and its relation to Lecture 1

Definition 5.1 (p. 80; slide 14). The pure-strategy profile $s^* = (s_1^*, \dots, s_n^*) \in S$ is a **Nash equilibrium** of G if s_i^* is a best response to s_{-i}^* for every $i \in I$:

$$u_i(s_i^*, s_{-i}^*) \;\geq\; u_i(s_i, s_{-i}^*) \qquad \forall s_i \in S_i,\ \forall i \in I. \tag{1}$$

In the slide's words: each player's strategy is a best response to the others' strategies, so **no player gains from a unilateral deviation**. In the Battle of the Sexes, (O, O) and (F, F) are Nash equilibria (slide 14).

Equivalently $s_i^* \in BR_i(s_{-i}^*)$ for every i: the profile is a **fixed point** of the best-response correspondences (this is how existence is proved, L02-09-existence).

> **The two requirements (p. 81):** (1) each player plays a best response to his beliefs; (2) the beliefs are correct. Requirement (1) is rationality; requirement (2) is the demanding one.

2b · C — how NE differs from iterated elimination (slide 15)

	IESDS and rationalizability	Nash equilibrium
Kind of concept	**epistemic**: follows from common knowledge of rationality	**strategic stability**: beliefs are **correct**

	IESDS and rationalizability	Nash equilibrium
Interpretations	what rational players who reason about each other can do	(i) **self-fulfilling expectations**; (ii) **self-enforcing agreements**; (iii) **social norms or conventions**

Gibbons's argument in §1 is interpretation (ii); Tadelis's influential Alex (§4) is (i); the side of the road (§2d) is (iii).

2c · C — the slides' two propositions, with their proofs (slides 16–17)

Notation for the proofs: S_i^k is player i's set of strategies that survive k rounds of IESDS ($S_i^0 = S_i$), and S_{-i}^k the rivals' surviving profiles.

Proposition 1 (slide 16). If s^* is a Nash equilibrium, then s^* survives IESDS.

Proof (by contradiction). 1. Suppose not, and let k be the **first** round in which some component s_i^* is eliminated. 2. Then s_i^* is strictly dominated by some $s_i' \in S_i^{k-1}$ against every $s_{-i} \in S_{-i}^{k-1}$. 3. No component of s^* was eliminated before round k, so $s_{-i}^* \in S_{-i}^{k-1}$. 4. Applying step 2 at $s_{-i} = s_{-i}^*$: $u_i(s_i', s_{-i}^*) > u_i(s_i^*, s_{-i}^*)$. 5. So s_i^* is not a best response to s_{-i}^*, contradicting (1). ■

> **Note (slide 16).** A Nash equilibrium **may involve weakly dominated strategies**; such equilibria are often considered less plausible. The lecture's example is Bertrand: $p_i^* = c$ is weakly dominated by any higher price at which demand is positive, yet (c, c) is the unique NE (L02-04-bertrand §0, slide 36). Proposition 1 is about **strict** dominance only.

Proposition 2 (slide 17). In a finite game, if s^* is the unique profile that survives IESDS, then s^* is a Nash equilibrium (and, by Proposition 1, the unique one).

Proof (by contradiction). 1. Suppose some player i has s_i with $u_i(s_i, s_{-i}^*) > u_i(s_i^*, s_{-i}^*)$. 2. Since $s_i \neq s_i^*$ and s^* is the unique survivor, s_i was eliminated in some round, by some s_i' that strictly dominates it against all surviving s_{-i}, **including** s_{-i}^* (never eliminated): $u_i(s_i', s_{-i}^*) > u_i(s_i, s_{-i}^*)$. 3. If s_i' is eliminated later, repeat the argument with s_i' in the role of s_i. 4. The game is finite, so the chain ends at a strategy that is never eliminated, that is, at s_i^*. 5. Payoffs against s_{-i}^* increase along the chain, so $u_i(s_i^*, s_{-i}^*) > u_i(s_i, s_{-i}^*)$, contradicting step 1. ■

These are Gibbons's Propositions B and A (Appendix 1.1.C, pp. 12–14) and the two halves of Tadelis's Prop. 5.1 below.

Proposition 5.1 (p. 80–81). If s^* is (1) a strict dominant-strategy equilibrium, (2) the unique survivor of IESDS, or (3) the unique rationalizable profile, then s^* is the **unique** Nash equilibrium.

Why, step by step. (a) s^* is a NE: by Prop. 4.4 (L02-01-best-response-rationalizability §2) each s_i^* is a best response to s_{-i}^*. (b) No other NE: a NE never contains a strictly dominated strategy (a strictly dominated strategy is never a best response, Prop. 4.3), and the same holds round by round in the reduced games, so **every NE survives IESDS** (Gibbons Appendix 1.1.C, Proposition B, pp. 12–13). If only s^* survives, s^* is the only candidate. (The proof is Tadelis exercise 5.1, p. 95.)

If the profile is…	then it is…	the converse
a strict dominant-strategy equilibrium	the unique IESDS survivor and the unique NE	fails: Cournot has a unique NE and no dominant strategy

If the profile is…	then it is…	the converse
the unique IESDS survivor	the unique NE	fails: the 3×3 game of §3 has a unique NE and nothing is eliminated
a NE	an IESDS survivor and rationalizable	fails: in the Battle of the Sexes (O, F) survives both and is not a NE

Worked (Tadelis p. 80, Lecture 1's IESDS game): rows U, M, D; columns L, C, R; (U, L) is the only IESDS survivor and the only NE, since $BR_1(L) = \{U\}$ and $BR_2(U) = \{L\}$. In the prisoner's dilemma the unique NE is (F, F): each plays a dominant strategy, which is a best reply to anything (slides 4–5: (F, F) is the strict dominant-strategy equilibrium, and slide 6: (M, M) Pareto dominates it).

2d · Worked — coordination and dilemmas (slide 18)

Which side of the road?

1 2	L	R
L	1, 1	0, 0
R	0, 0	1, 1

Best replies: to L it is L ($1 > 0$), to R it is R ($1 > 0$), for both players. Mutual best replies: **(L, L) and (R, R)**. Nothing is dominated (L is better against L, worse against R), so **IESDS eliminates nothing**: the equilibrium concept does the selecting, and which side a country drives on is a **convention** (interpretation iii).

Prisoner's dilemma (slide 18 payoffs).

1 2	C	D
C	2, 2	−1, 3
D	3, −1	0, 0

Player 1: against C, D gives $3 > 2$; against D, D gives $0 > -1$. D strictly dominates C, and by symmetry for player 2. **Unique NE (D, D)**, although (C, C) gives $2 > 0$ to both and so Pareto dominates it.

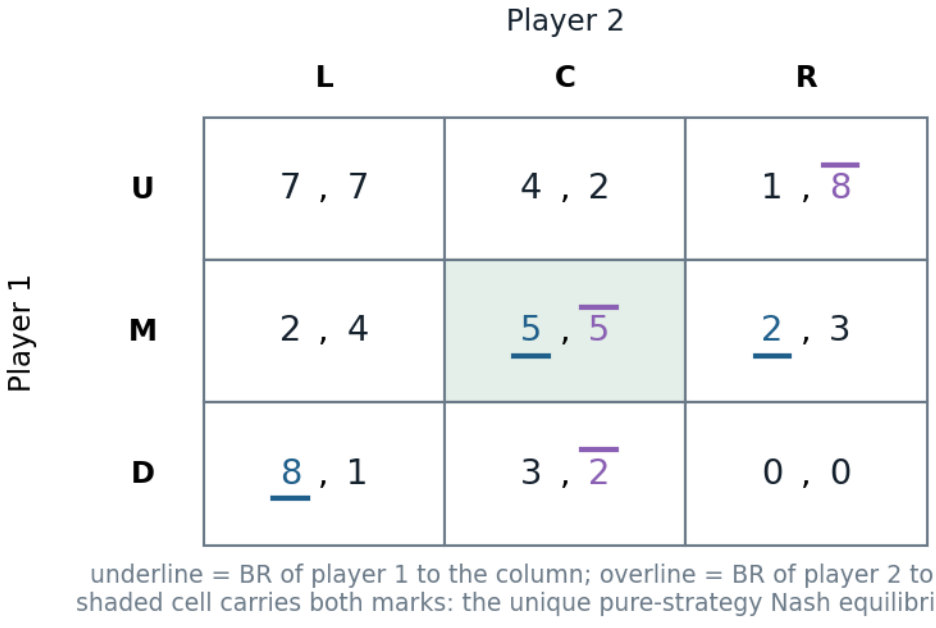
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3 · C — finding pure equilibria in a matrix: the underline method (§5.1.1, pp. 81–82)

The game of p. 81, where **nothing is dominated**, so IESDS and rationalizability say "anything can happen":

P1 P2	L	C	R
U	7, 7	4, 2	1, 8
M	2, 4	5, 5	2, 3
D	8, 1	3, 2	0, 0

- **Step 1 (player 1, column by column).** L: max(7, 2, 8) = 8 → D. C: max(4, 5, 3) = 5 → M. R: max(1, 2, 0) = 2 → M. Player 1 best-responds at (D, L), (M, C), (M, R).
- **Step 2 (player 2, row by row).** U: max(7, 2, 8) = 8 → R. M: max(4, 5, 3) = 5 → C. D: max(1, 2, 0) = 2 → C. Player 2 best-responds at (U, R), (M, C), (D, C).
- **Step 3.** The only cell marked twice is **(M, C)**: the unique pure-strategy NE.



Underline = player 1's best reply to the column; overline = player 2's best reply to the row; the shaded cell (M, C) is the unique pure NE. Source: Tadelis §5.1.1, pp. 81–82.

Note what the figure shows that IESDS could not: every row and column is a best reply to something, so nothing is eliminated, yet only one cell is a **mutual** best reply.

> **"Unique pure-strategy NE" is the exact claim (p. 82).** The game also has **four mixed equilibria** (support enumeration, L02-12-python §7): for instance both players put $\frac{1}{6}$ on their first and $\frac{5}{6}$ on their second strategy, which makes U and M pay player 1 the same 4.5 (and D only $\frac{23}{6}$), and likewise for player 2. The underline method finds pure equilibria only. Gibbons uses the same device (Figure 1.1.5, p. 10) and so does Cabral, marking best replies with asterisks (Fig. 7.5, pp. 165–166).
>
> **Interactive companion.** The Mixing Square (`companions/companion-mixed-2x2.html`) — its matrix panel applies the underline method live to any 2×2 game; when no cell is marked twice the equilibrium must be mixed.

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4 · C — multiple equilibria and what selects among them

Battle of the Sexes (pp. 79, 81, 83). (O, O) and (F, F) are both NE; (O, F) and (F, O) are not (each player would move to the other's event). Tadelis's selection story: if Alex is *influential*, the belief that both go to the opera is **self-fulfilling**, not because Chris likes pleasing Alex (that would change the payoffs) but because beliefs coordinate. As the outside analyst, "one of these two outcomes is what we predict" (p. 81).

Stag Hunt (§5.2.1, pp. 83–84). Not covered in class: reading; the slides' coordination example is

the road game (§2d).

1 2	S	H
S	5, 5	0, 3
H	3, 0	3, 3

Best replies: to S the best reply is S ($5 > 3$); to H it is H ($3 > 0$). Both (S, S) and (H, H) are NE, and (S, S) **Pareto dominates** (H, H). Yet (H, H) is a reasonable prediction: a hunter who expects the other to hunt hare gets 0 from the stag alone. Tadelis's reading: identical societies end up cooperative or individualist through **self-fulfilling beliefs, or norms** (p. 84; the "big push" in footnote 2).

Game	NE (pure)	Selection argument
Battle of the Sexes	(O, O), (F, F)	influence, history, convention
Stag Hunt	(S, S), (H, H)	Pareto dominance vs the safety of H (risk)
Cabral Fig. 7.6 (standards)	two compatible standards	**focal points** (Cabral pp. 166–167)

Cabral's "meet me in New York" example (p. 167): infinitely many NE, yet most subjects choose Grand Central at noon, a **focal point** that comes from outside the game.

> **Multiplicity is not a failure of the concept.** Nash equilibrium is a consistency condition; which consistent profile is played is a question about beliefs that the payoff matrix alone cannot answer (rules/game-theory items 3 and 17).

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5 · C — evaluating Nash equilibrium (§5.1.2, p. 83)

Criterion	Verdict
Existence	not always in pure strategies (Matching Pennies, L02-07-mixed-strategies; asymmetric Bertrand with continuous prices, L02-04-bertrand); always in mixed strategies for finite games (L02-09-existence)
Uniqueness	often fails (above); but sharper than IESDS and rationalizability, by Prop. 5.1
Pareto	**not guaranteed**: prisoner's dilemma, the tragedy of the commons (L02-05-commons)

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6 · G — MWG §8.D (pp. 246–249)

MWG Def. 8.D.1 (p. 246) states the pure-strategy NE exactly as Def. 5.1. MWG discusses why players might play a NE (pp. 248–249): a self-enforcing agreement reached by pre-play communication, a stable social convention, a focal point, or the outcome of learning; it also stresses that common knowledge of

rationality alone does **not** imply NE play, it implies only rationalizability. Named here; not examinable unless the lecture states it.

Links

- Neighbours: L02-01-best-response-rationalizability, L02-03-cournot, L02-07-mixed-strategies.
- Lecture 1 (dominance, IESDS, Pareto): Lecture 1; Lecture 3 refines NE with subgame perfection: Lecture 3.
- Rules pack: rules/game-theory.
- Practice: Tadelis 5.1, 5.2, 5.3 (p. 95), 5.4–5.6 (p. 96); Gibbons 1.1 (p. 48); Muñoz-Garcia & Toro-Gonzalez Ch. 2 (pp. 37–84).

Slides 20–27 · 5.2.3 · Cournot duopoly: two best-response lines and one crossing

Claim. With continuous strategies a Nash equilibrium is found by **solving every player's first-order condition at once**: each FOC is a best-response function, and the equilibrium is where they cross. In Cournot the lines slope **down** (strategic substitutes) and cross once, at $q_i^* = (a - 2c_i + c_j)/(3b)$. The duopoly produces **more than a monopoly and less than competition**, because **collusion at the monopoly output is not a best reply to itself**.

Prev: L02-02-nash-equilibrium · Next: L02-04-bertrand · Map: L02-00-map

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0 · C — the lecture's version (slides 20–24), in its notation

Notation. $q_i \geq 0$ firm i's output; $Q = q_1 + q_2$; inverse demand $P(Q) = a - bQ$ ($a > c \geq 0$, $b > 0$); constant marginal cost c for both firms; Π_i firm i's profit; q^m the monopoly output; q^{cp} the perfectly competitive output. Example 2 (slide 20): $a = 100$, $b = 1$, $c = 10$.

The game (slide 21). Players $I = \{1, 2\}$; strategies $S_i = \{q_i : q_i \geq 0\}$; payoffs

$$\Pi_i(q_i, q_j) = P(q_i + q_j)\, q_i - c\, q_i = (a - c - b(q_i + q_j))\, q_i. \tag{1}$$

Example: $\Pi_i = (100 - q_i - q_j)q_i - 10q_i$. Slide 20 asks for (a) this normal form, (b) dominant and dominated strategies, (c) the best responses and (d) the equilibrium. For (b): no output is dominant, because the best output depends on q_j (by (5) below); outputs above q^m are strictly dominated, which starts the IESDS sequence below.

Two benchmarks (slide 22). A monopolist sets marginal revenue equal to cost: $RM(q^m) = a - 2bq^m = c$, so

$$q^m = \frac{a - c}{2b}. \tag{2}$$

Many identical firms push the price to cost: $P(q^{cp}) = a - bq^{cp} = c$, so

$$q^{cp} = \frac{a - c}{b} = 2q^m. \tag{3}$$

Example: $q^m = \dfrac{100 - 10}{2} = 45$ and $q^{cp} = 90$.

Best response (slide 23). Given q_j, firm i maximizes (1). First-order condition:

$$\frac{\partial \Pi_i}{\partial q_i} = a - c - 2bq_i - bq_j = 0. \tag{4}$$

Solving (4) for q_i: $q_i = \dfrac{a - c - bq_j}{2b} = \dfrac{a - c}{2b} - \dfrac{1}{2}q_j$, and by (2)

$$q_i^*(q_j) = \begin{cases} q^m - \frac{1}{2}q_j & 0 \leq q_j < q^{cp}, \\ 0 & q_j \geq q^{cp}. \end{cases} \tag{5}$$

The corner: $q^m - \frac{1}{2}q_j \leq 0$ exactly when $q_j \geq 2q^m = q^{cp}$ by (3). Example: $100 - 2q_i - q_j - 10 = 0$, so $q_i^*(q_j) = \dfrac{90 - q_j}{2}$.

Nash equilibrium (slide 24). A pair of mutual best responses: $q_1 = q^m - \frac{1}{2}q_2$ and $q_2 = q^m - \frac{1}{2}q_1$. Substituting the second into the first:

$$q_1 = q^m - \tfrac{1}{2}(q^m - \tfrac{1}{2}q_1)$$

$$q_1 = \tfrac{1}{2}q^m + \tfrac{1}{4}q_1$$

$$\tfrac{3}{4}q_1 = \tfrac{1}{2}q^m \implies q_1^* = \tfrac{2}{3}q^m,$$

and by symmetry

$$q_1^* = q_2^* = \tfrac{2}{3}q^m. \tag{6}$$

Example: $q^* = \frac{2}{3}(45) = 30$, $Q^* = 60$, $P^* = 100 - 60 = 40$, $\Pi^* = (40-10)(30) = 900$.

IESDS gives the same answer (slide 24). By (5) the largest best response is q^m (at $q_j = 0$), so outputs above q^m are never optimal; given $q_j \le q^m$, the smallest best response is $q^m - \frac{1}{2}q^m = \frac{1}{2}q^m$; and so on. The surviving interval's end points alternate:

$$\tfrac{1}{2}q^m, \quad q^m - \tfrac{1}{2}\cdot\tfrac{1}{2}q^m = \tfrac{3}{4}q^m, \quad q^m - \tfrac{1}{2}\cdot\tfrac{3}{4}q^m = \tfrac{5}{8}q^m, \quad q^m - \tfrac{1}{2}\cdot\tfrac{5}{8}q^m = \tfrac{11}{16}q^m, \ \ldots \ \to \ \tfrac{2}{3}q^m.$$

Example: 22.5, 33.75, 28.125, 30.9375, … → 30 (the rounds of L02-01-best-response-rationalizability §3).

Market structure	Total output	Example
Monopoly (or cartel)	q^m	45
Cournot duopoly	$\frac{4}{3}q^m$	**60**
Perfect competition	$q^{cp} = 2q^m$	90

The rest of this note is Tadelis's version with **different** marginal costs c_i; set $c_1 = c_2 = c$ and it collapses to (5)–(6).

1 · C — the game and the best response (Tadelis p. 87)

Players: firms 1, 2. Strategies: $q_i \ge 0$. Inverse demand $P = a - b(q_1 + q_2)$, costs $c_i(q_i) = c_i q_i$, payoffs

$$u_i(q_i, q_j) = (a - bq_i - bq_j)q_i - c_i q_i.$$

First-order condition, given a belief q_j:

$$\frac{\partial u_i}{\partial q_i} = a - 2bq_i - bq_j - c_i = 0.$$

Second-order condition: $\partial^2 u_i / \partial q_i^2 = -2b < 0$, so the FOC gives the maximum. Solving for q_i, and truncating at zero (a firm never produces at a loss),

$$\boxed{BR_i(q_j) = \max\left\{0, \ \frac{a - bq_j - c_i}{2b}\right\}} \qquad \text{(eq. 5.2)}$$

Slope $-1/2$: one more unit from the rival cuts the best reply by half a unit. **Strategic substitutes** (Tadelis's remark, p. 93): a best reply that falls in the rival's choice.

2 · C — solving the two lines (every step)

Interior equilibrium: both firms on their lines,

$$2bq_1 + bq_2 = a - c_1, \qquad bq_1 + 2bq_2 = a - c_2.$$

Multiply the first by 2 and subtract the second: $3bq_1 = 2(a-c_1)-(a-c_2) = a-2c_1+c_2$. Symmetrically for firm 2:

$$\boxed{q_1^* = \frac{a-2c_1+c_2}{3b}, \qquad q_2^* = \frac{a-2c_2+c_1}{3b}}$$

Then $Q^* = \dfrac{2a-c_1-c_2}{3b}$ and $P^* = a - bQ^* = \dfrac{a+c_1+c_2}{3}$.

Profits. From the FOC, $P^* - c_i = bq_i^*$ (substitute $a - bq_j^* - c_i = 2bq_i^*$ into $P^* - c_i = a - bq_i^* - bq_j^* - c_i$), so

$$\pi_i^* = (P^* - c_i)q_i^* = b\,(q_i^*)^2.$$

Corner. If $a - 2c_1 + c_2 < 0$, i.e. $c_1 > (a + c_2)/2$, the formula gives $q_1^* < 0$. Then firm 1 produces 0 and firm 2 is a monopolist, $q_2 = (a - c_2)/(2b)$. Check it is an equilibrium: $BR_1((a - c_2)/(2b)) = \max\{0, (a - 2c_1 + c_2)/(4b)\} = 0$. ✓

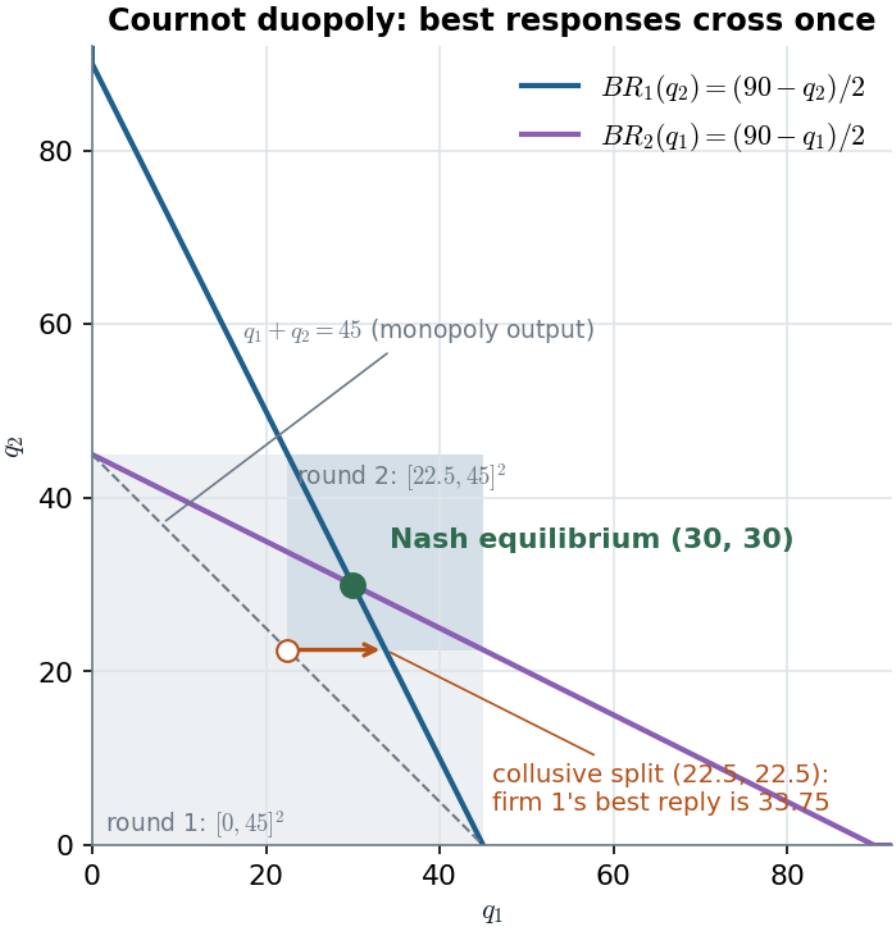
Comparative statics (interior). $\partial q_1^*/\partial c_1 = -2/(3b)$, $\partial q_2^*/\partial c_1 = +1/(3b)$, $\partial Q^*/\partial c_1 = -1/(3b)$, $\partial P^*/\partial c_1 = +1/3$. A cost increase is passed to the price only one-third, and the rival picks up half of the lost output.

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3 · Worked — Tadelis's numbers ($a = 100$, $b = 1$, $c_1 = c_2 = 10$; pp. 87–88, 90)

Quantity	Computation	Value
$BR_i(q_j)$	$(100 - q_j - 10)/2$	$(90 - q_j)/2$
q_i^*	$(100 - 20 + 10)/3$	**30**
Q^*, P^*	60; $100 - 60$	**60, 40**
π_i^*	$(40 - 10) \cdot 30$	**900**
Monopoly	$\max(100 - q)(q) - 10q$: $90 - 2q = 0$	$q = 45$, $P = 55$, $\pi = 2025$
Competitive	$P = c = 10$	$Q = 90$
Joint duopoly profit	$2 \cdot 900$	$1800 < 2025$

$$\underbrace{45}_{\text{monopoly}} < \underbrace{60}_{\text{Cournot}} < \underbrace{90}_{\text{competitive}}, \qquad Q^*/Q^c = \tfrac{2}{3}.$$



The two lines of eq. (5.2) cross once at (30, 30). The hollow dot is the collusive split (22.5, 22.5) on the monopoly line; the arrow is firm 1's best reply to 22.5, which is 33.75. Source: Tadelis Fig. 5.2, p. 88; Gibbons p. 17.

The arrow is the whole economics of the figure: starting from the split that maximizes joint profit, each firm wants to expand, and only at the crossing does no firm want to move.

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4 · U — why collusion fails (Gibbons p. 17)

Joint profit is maximal at total output $q_m = (a - c)/(2b)$, e.g. $q_i = q_m/2$ each. But

$$BR_i\left(\frac{q_m}{2}\right) = \frac{a - c - b\,q_m/2}{2b} = \frac{a - c - (a - c)/4}{2b} = \frac{3(a - c)}{8b} > \frac{a - c}{4b} = \frac{q_m}{2}.$$

With Tadelis's numbers: $BR_i(22.5) = (90 - 22.5)/2 = 33.75 > 22.5$. Deviating raises firm i's profit from $(55 - 10) \cdot 22.5 = 1012.5$ to $(100 - 56.25 - 10) \cdot 33.75 = 1139.06$.

> **The low monopoly output keeps the price high, and at a high price each firm wants to sell more even though doing so drives the price down.** In equilibrium the price has fallen just enough that the temptation to expand is exactly offset (Gibbons p. 17). This is the prisoner's-dilemma structure of oligopoly that Lectures 5–6 will try to escape with repetition (Lecture 6).

4b · C — the cartel as a prisoner's dilemma (slide 25)

A **cartel**: competing firms coordinate to maximize joint profit, as a single monopolist. With identical costs each produces $q^m/2$, 22.5 in the example. Restrict each firm to two choices, **Cartel** (22.5) or **Deviate** (the best response to the cartel output, $q_i^*(22.5) = \frac{90-22.5}{2} = 33.75$). The payoffs, worked:

Profile	Q	P	Profits
(Cartel, Cartel)	45	55	$(55-10)(22.5) =$ 1,012.5 each
(Deviate, Cartel)	56.25	43.75	deviator $(33.75)(33.75) =$ 1,139.06; cartel firm $(33.75)(22.5) = 759.38$

The slide's matrix (payoffs rounded):

Firm 1 \ Firm 2	Cartel	Deviate
Cartel	1,012, 1,012	759, 1,139
Deviate	1,139, 759	900, 900

> **Reading the (Deviate, Deviate) cell.** 900 is each firm's profit at the **Nash outputs** (30, 30), where mutual deviation ends up. If both literally produced 33.75, $Q = 67.5$, $P = 32.5$ and each would earn $(32.5-10)(33.75) = 759.38$, exactly what the cartel firm earns against a deviator; Deviate would then be only weakly dominant. With the slide's 900, Deviate strictly dominates Cartel (1,139 > 1,012 and 900 > 759).

"Which game has the same structure?" **The prisoner's dilemma**: each firm has a dominant strategy (deviate), and the equilibrium (900 each) is Pareto dominated by the cartel (1,012 each). Lectures 5–6 ask whether repetition can sustain the cartel (Lecture 6).

4c · C — Cournot oligopoly with n firms (slides 26–27)

Price $P(q_1 + \cdots + q_n) = a - b(q_1 + \cdots + q_n)$; firm i's payoff $q_i(P(\sum_j q_j) - c)$. The same FOC as (4), with $\sum_{j \neq i} q_j$ in place of q_j, gives

$$q_i^*(q_{-i}) = q^m - \tfrac{1}{2}\sum_{j \neq i} q_j \quad \text{(if positive)}. \tag{7}$$

In equilibrium (7) holds for every i at q^*. Multiply by 2 and move the sum:

$$2q_i^* = 2q^m - \sum_{j \neq i} q_j^* \implies q_i^* = 2q^m - \sum_{j \in I} q_j^* = 2q^m - Q^*. \tag{8}$$

Summing (8) over the n firms:

$$Q^* = 2nq^m - nQ^* \implies Q^* = \frac{2n}{n+1}\, q^m. \tag{9}$$

Substituting (9) back into (8):

$$q_i^* = 2q^m - \frac{2n}{n+1} q^m = \frac{2q^m}{n+1} \qquad (i = 1, \ldots, n). \tag{10}$$

Check: $n = 1$ gives q^m (monopoly); $n = 2$ gives $\frac{2}{3}q^m$, as in (6). As $n \to \infty$, $Q^* \to 2q^m = q^{cp}$: **with many identical firms the market approaches perfect competition** (slide 27). In Tadelis's notation (10) is $q_i^* = (a-c)/((n+1)b)$ and $P^* = (a+nc)/(n+1) \to c$ (Tadelis exercise 5.8, p. 96; Gibbons problem 1.4, p. 49). Cournot firms keep some market power when they are few; Bertrand firms lose it with two (L02-04-bertrand).

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5 · Asymmetric costs, worked (not on the slides; reading)

$a = 100$, $b = 1$, $c_1 = 1$, $c_2 = 2$ (the costs Tadelis uses for asymmetric Bertrand, p. 91):

	Formula	Value
q_1^*	$(100 - 2 + 2)/3$	33.333
q_2^*	$(100 - 4 + 1)/3$	32.333
P^*	$(100 + 1 + 2)/3$	34.333
$\pi_1^* = b(q_1^*)^2$	33.333^2	1111.11
π_2^*	32.333^2	1045.44

The cheaper firm sells exactly $(c_2 - c_1)/b = 1$ unit more, and **both** firms survive, unlike asymmetric Bertrand where the efficient firm takes the whole market.

> **Interactive companion.** The Cournot Crossing (`companions/companion-cournot-best-responses.html`) — raise c_1 to the exit threshold $(a + c_2)/2$ (55 at the defaults; there is a button) and watch firm 1's line slide inside firm 2's until the crossing reaches the axis; change b and see every quantity rescale while prices stay put.

Links

- Neighbours: L02-01-best-response-rationalizability §3 (the same game by rationalizability), L02-04-bertrand (prices instead of quantities), L02-05-commons (another strategic-substitutes game).
- Lecture 1 (Cournot by IESDS, Tadelis p. 65): Lecture 1; Lecture 3 (Stackelberg, the sequential version): Lecture 3; Lecture 6 (Cournot collusion under repetition): Lecture 6.
- Code: L02-12-python §3; the Cournot IESDS recursion of Lecture 1 lives in `Listas/lista-lecture-01-check.py` (`BR`, `cournot_iesds`).
- Practice: Tadelis 5.8 (p. 96), 5.11 (p. 97); Gibbons 1.4, 1.5 (p. 49), 1.6 (pp. 49–50); Cabral §7.1 (pp. 161–167).

Slides 29–40 · 5.2.4 · Bertrand duopoly: two firms are enough for marginal-cost pricing

Claim. Change the strategy from quantity to price and the same market behaves competitively: with identical costs the unique equilibrium of the continuous game is $p_1 = p_2 = c$ (Prop. 5.2), proved by showing every other profile has a profitable undercut. On a one-cent grid there are exactly **two** equilibria, (10, 10) and (10.01, 10.01). With **unequal** costs and continuous prices **no pure equilibrium exists**: the payoff jumps at the rival's price, so the efficient firm has no best response. Differentiated products restore interior, upward-sloping best responses (**strategic complements**).

Prev: L02-03-cournot · Next: L02-05-commons · Map: L02-00-map

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0 · C — the lecture's version (slides 29–40), in its notation

Notation. $p_i \geq 0$ firm i's price; $c \geq 0$ the common marginal cost; market demand at the lowest price p is $Q(p) = \frac{a}{b} - \frac{p}{b}$ for $p < a$ and 0 otherwise (the inverse of $P = a - bQ$); p^m the monopoly price.

The game (slide 29). Two firms sell a homogeneous good and choose prices simultaneously; consumers buy from the cheapest firm and split equally at a tie:

$$u_i(p_i, p_j) = \begin{cases} (p_i - c)\, Q(p_i) & p_i < p_j, \\ \frac{1}{2}(p_i - c)\, Q(p_i) & p_i = p_j, \\ 0 & p_i > p_j. \end{cases} \tag{1}$$

The monopoly price (slide 30). Maximize $(p-c)Q(p) = (p-c)\dfrac{a-p}{b}$. FOC:

$$\frac{(a-p)-(p-c)}{b} = 0 \implies p^m = \frac{a+c}{2}. \tag{2}$$

(With Tadelis's numbers $a = 100$, $c = 10$: $p^m = 55$, as in §1 below.)

Firm i's payoff given p_j, case by case (slides 31–33).

Rival's price	What firm i can earn	Best response
$p_j < c$	any $p_i \leq p_j$ sells at a loss; any $p_i > p_j$ sells nothing and earns 0	every $p_i > p_j$
$p_j = c$	$p_i < c$ loses; $p_i = c$ and $p_i > c$ both earn 0	every $p_i \geq p_j$
$c < p_j \leq p^m$	i wants $p_i < p_j$ **as close as possible** to p_j; for any such p_i a higher price still below p_j earns more	**none**: the existence problem
$p^m < p_j$	any price below p_j takes the whole market, and the best of them is p^m	p^m

Best response (slide 34).

$$p_i^*(p_j) = \begin{cases} \{p_i : p_i > p_j\} & p_j < c, \\ \{p_i : p_i \geq p_j\} & p_j = c, \\ \emptyset & c < p_j \leq p^m, \\ \{p^m\} & p^m < p_j. \end{cases} \tag{3}$$

(c, c) **is a Nash equilibrium (slide 35).** Suppose $p_2 = c$. By (3) any $p_1 \geq c$ is a best response; in particular c is. The same argument holds for firm 2.

> **A weakly dominated equilibrium strategy (slide 36).** $p_i^* = c$ is weakly dominated by any higher price at which demand is positive: such a price earns 0 when undercut and a strictly positive profit when it is the lowest or tied, while c earns 0 always. This is slide 16's caveat in action (L02-02-nash-equilibrium §2c).

(c, c) **is the unique equilibrium (slides 37–38).**

Step 1. In any NE, $p_1^* = p_2^*$. Suppose $p_1^* > p_2^*$. Three cases: - $p_2^* < c$: firm 2 makes a loss, when it could earn zero with any $p_2' > p_1^*$; - $p_2^* = c$: firm 2 earns zero, when it could earn a positive profit with $p_2' = p_1^*$ (half the market at a margin $p_1^* - c > 0$); - $p_2^* > c$: firm 1 earns zero, when it could earn a positive profit with any $p_1' \in (c, p_2^*)$.

An analogous argument rules out $p_1^* < p_2^*$.

Step 2. In any NE, $p_1^* = p_2^* = c$. - If $p_1^* = p_2^* > c$, firm 1 could charge $p_1' \in (c, p_2^*)$ close to p_2^* and serve the whole market instead of sharing it (almost twice the profit). - If $p_1^* = p_2^* < c$, both make losses, and each would rather raise its price and sell nothing.

"According to the model, a duopoly in Nash equilibrium works like a competitive market" (slide 38).

The Bertrand paradox (slide 39). Two firms behave as if in perfect competition: price equals marginal cost and profits are zero. The assumptions behind it: (i) **homogeneous products**; (ii) **simultaneous choices**; (iii) **unlimited production capacity**. Relax any one and prices rise above cost (§5 relaxes (i)).

Cournot or Bertrand? (slide 40). It depends on what is harder to adjust.

Model	Harder to adjust	Strategic decision	Slide's examples
Cournot	quantity (capacity)	how much to produce	cars, cement, steel, commodities
Bertrand	price, or quantity expands at almost no cost	the price	software and other digital goods

In both models the NE predicts **more output than a cartel**, but the Bertrand prediction is much more extreme ($Q^* = q^{cp}$ against Cournot's $\frac{4}{3}q^m$).

The sections below are Tadelis's treatment. §1 and §3 are the same result in his numbers; **§2 (cent grid), §4 (unequal costs) and §5 (differentiated products) were not covered in class** and are reading.

1 · C — the game (p. 88)

Demand $p = 100 - q$, $c_i(q_i) = 10q_i$, prices $p_i \in [0, \infty)$; the lower price takes the market, a tie splits it:

$$u_i(p_i, p_j) = \begin{cases} (100 - p_i)(p_i - 10) & p_i < p_j \\ \frac{1}{2}(100 - p_i)(p_i - 10) & p_i = p_j \\ 0 & p_i > p_j \end{cases}$$

Monopoly benchmark (p. 89). $\max_p (100 - p)(p - 10)$: FOC $110 - 2p = 0$, so $p = 55$, $q = 45$, profit $45 \cdot 45 = 2025$.

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2 · Not covered in class — the best response on a cent grid (pp. 89–90)

Prices in steps of $\Delta = 0.01$. Go case by case over the rival's price p_j:

p_j	Best response	Why
$p_j > 55$	55	the monopoly price already undercuts p_j
$10.02 \le p_j \le 55$	$p_j - 0.01$	undercutting by a cent beats tying (below)
$p_j = 10.01$	10.01	undercutting to 10 earns 0; tying earns $\frac{1}{2}(89.99)(0.01) > 0$
$p_j = 10$	any $p_i \ge 10$	every such price earns exactly 0
$p_j < 10$	any $p_i > p_j$	matching or undercutting sells below cost

The undercut beats the tie, every step (p. 89). Let Δv be the gain from $p_j - 0.01$ over p_j:

$$\Delta v = (100 - (p_j - 0.01))((p_j - 0.01) - 10) - \tfrac{1}{2}(100 - p_j)(p_j - 10).$$

Expand the first product: $(100.01 - p_j)(p_j - 10.01) = -p_j^2 + 110.02p_j - 1001.1001$. The second: $\frac{1}{2}(-p_j^2 + 110p_j - 1000) = -0.5p_j^2 + 55p_j - 500$. Subtract:

$$\Delta v = -0.5p_j^2 + 55.02p_j - 501.1001 \approx 55.02p_j - 0.5p_j^2 - 501.1.$$

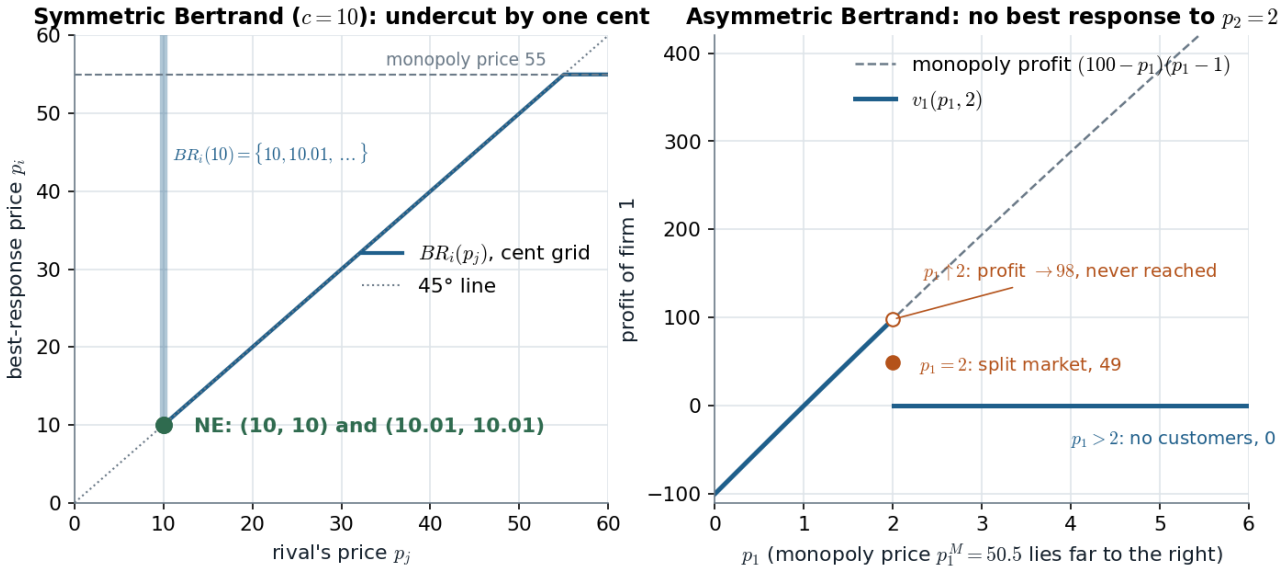
At $p_j = 10.02$ the book's rounded form gives $551.3004 - 50.2002 - 501.1 = 0.0002 > 0$; the exact constant gives 0.0001, which is also the direct comparison: undercutting to 10.01 earns $89.99 \times 0.01 = 0.8999$, tying at 10.02 earns $\frac{1}{2} \times 89.98 \times 0.02 = 0.8998$. Either way it is positive. And $d\Delta v/dp_j = 55.02 - p_j > 0$ for $p_j < 55.02$, so $\Delta v > 0$ on all of $[10.02, 55]$.

Equilibria. Look for mutual best replies: $BR_i(10.01) = \{10.01\}$ and $10 \in BR_i(10)$, so

$$(p_1, p_2) \in \{(10, 10),\ (10.01, 10.01)\}.$$

No other pair works: at or above 10.02 the higher-priced (or tied) firm undercuts; $(10, p)$ with $p > 10$ fails because the firm at **10** earns zero and would rather raise its price to just below p for a positive profit.

> **Tadelis's warning (p. 90).** When $p_2 = 10$, *every* $p_1 \geq 10$ is a best reply for firm 1. It does **not** follow that $(p_1, 10)$ is an equilibrium for every $p_1 \geq 10$: if $p_1 > 10$, firm 2's price of 10 is no longer a best reply to p_1. **An equilibrium needs both players best-responding, not one.**



Left: the cent-grid best response, computed by brute force; the shaded vertical band is $BR_i(10) = \{10, 10.01, \ldots\}$. *Right: Tadelis's Fig. 5.3 with the book's numbers* ($c_1 = 1$, $p_2 = 2$)*: the supremum 98 is approached but never attained. Source: pp. 88–93.*

On the left, the best response hugs the 45° line one cent below it, so the only places it can meet its mirror image are where undercutting stops paying: at 10 and 10.01. On the right, the open circle is the whole existence problem.

Profits compared (p. 90). Cournot: 900 each. Bertrand: 0 at (10, 10) or about 0.45 at (10.01, 10.01). Monopoly: 2025. Adding a third or fourth Bertrand firm changes nothing.

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3 · C — Proposition 5.2: continuous prices (p. 91; the slides' result, slides 35–38)

Proposition 5.2. With $\Delta = 0$, $c_i(q_i) = cq_i$ for both firms and $p = a - bq$, $a > c$, the unique Nash equilibrium is $p_1 = p_2 = c$.

Proof, every case. 1. No equilibrium has $p_i < c$: the firm with the lower price sells at a loss and would rather raise its price. So $p_i \geq c$ for both. 2. $p_1 = p_2 = \hat{p} > c$: each earns half the market at margin $\hat{p} - c$; cutting to $\hat{p} - \varepsilon$ takes the whole market at almost the same margin, nearly doubling profit. Not an equilibrium. 3. $p_1 > p_2 > c$ (or the mirror): firm 2 earns the whole market but can raise its price to $p_1 - \varepsilon$ and earn more (as long as p_2 is below the monopoly price; above it, firm 1 undercuts). Not an equilibrium. 4. $p_1 > p_2 = c$: firm 2 earns zero and can raise its price toward p_1 for a positive profit. Not an equilibrium. 5. $p_1 = p_2 = c$: both earn zero; any $p_i > c$ sells nothing (zero), any $p_i < c$ loses. Every price $\geq c$ is a best reply to c, so c is one. **Equilibrium.** ■

Two firms reproduce perfect competition. "One firm may have monopoly power, but when we let one more firm compete, and they compete on prices, then the market will behave competitively" (p. 90).

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4 · Not covered in class — unequal costs: no equilibrium in pure strategies (pp. 91–93)

$c_1 = 1$, $c_2 = 2$, $p = 100 - q$.

With cents. $(p_1, p_2) = (1.99, 2.00)$ is an equilibrium: firm 1 takes the market at a margin of 0.99 and would lose by raising to 2.00 (split) or above (nothing); firm 2 sells nothing and would lose money by pricing below 2. (Tadelis exercise 5.12, p. 98, asks for the others.)

Continuous prices. The "limit" $p_1 = p_2 = 2$ is **not** an equilibrium: firm 1 earns $\frac{1}{2}(98)(1) = 49$ and would earn almost 98 by pricing a hair below 2. But there is no best "hair":

$$u_1(p_1, 2) = \begin{cases} (100 - p_1)(p_1 - 1) & p_1 < 2 \quad (\to 98) \\ 49 & p_1 = 2 \\ 0 & p_1 > 2 \end{cases}$$

The supremum 98 is **not attained**: for any $p_1 < 2$, a price closer to 2 earns more. Firm 1 has **no best response** to $p_2 = 2$, nor to any $p_2 < 50.5$ (its monopoly price, footnote 7, p. 92).

Ruling out every candidate (pp. 92–93). $p_i \geq p_j > 2$: firm i deviates to a price in $(2, p_j)$. $p_i \geq p_j < 1$: firm j sells at a loss and deviates up. So any equilibrium has prices in $[1, 2]$; there, $p_1 \geq p_2$ fails (firm 1 undercuts just below p_2) and $p_2 \geq p_1$ fails (firm 2 prefers any price above p_1). **No pure equilibrium.**

The culprit is a discontinuous payoff, not the solution concept. Nash's existence theorem (L02-09-existence) needs finite games (or continuous payoffs); this game has neither. Tadelis's pragmatic line (p. 93): discreteness removes the problem, so continuous prices are a convenience to be dropped when they create such anomalies.

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5 · Not covered in class — strategic complements and differentiated products

Tadelis's Remark (p. 93). Cournot best replies fall in the rival's choice (**strategic substitutes**; also the commons). Bertrand best replies **rise** with the rival's price between marginal cost and the monopoly price (**strategic complements**). Note a slip in the printed text: the Remark says the monopoly price is "equal to 45"; in the leading example it is **55** (45 is the monopoly *quantity*, p. 89).

Gibbons §1.2.B (pp. 21–22): differentiated products. Demand $q_i = a - p_i + bp_j$ with $0 < b < 2$, marginal cost $c < a$. Profit $\pi_i = (a - p_i + bp_j)(p_i - c)$.

$$\frac{\partial \pi_i}{\partial p_i} = (a - p_i + bp_j) - (p_i - c) = 0 \implies p_i = \frac{a + bp_j + c}{2}.$$

Slope $b/2 > 0$: **upward-sloping** best responses. Symmetric solution $p^* = \frac{a + bp^* + c}{2} \Rightarrow (2 - b)p^* = a + c$:

$$\boxed{p_1^* = p_2^* = \frac{a+c}{2-b}}$$

which is why the model needs $b < 2$. Prices are above cost: product differentiation softens price competition. Gibbons stresses that "Cournot equilibrium" and "Bertrand equilibrium" are **the same concept (Nash) applied to two different games** (p. 21).

	Cournot (quantities)	Bertrand, homogeneous	Bertrand, differentiated
Best response slope	$-1/2$ (substitutes)	$+1$ one cent below (complements)	$+b/2$ (complements)
Price, $a = 100$, $c = 10$	40	10	$(a+c)/(2-b)$
Market power with 2 firms	partial	none	partial

Links

- Neighbours: L02-03-cournot (quantities), L02-09-existence (why discontinuity breaks existence).
- Applications: Cabral §7.1 (pp. 161–167) for the IO setting; Lecture 6 revisits Bertrand collusion under repetition (Lecture 6).
- Code: L02-12-python §4 (the cent-grid best response, brute force).
- Practice: Tadelis 5.12 (p. 98), 5.16 (p. 99); Gibbons 1.7 (pp. 49–50).

5.2.2 · The tragedy of the commons: an equilibrium everyone would vote to leave

Claim. When each player's use of a shared resource harms **all** users but the user counts only **his own** share of the harm, the Nash equilibrium over-uses the resource. In Tadelis's clean-air game each of two players takes $K/3$ where $K/4$ would make **both** better off; in Gibbons's village too many goats graze because the equilibrium condition weighs the harm by G^*/n instead of G^{**}. **Nash equilibrium is not Pareto optimal.**

Not covered in class. The Lecture 2 slides (`Aula/Lecture02.pdf`) do not teach the tragedy of the commons. This note is kept as reading and practice; it is examinable only if a problem set brings it.

Prev: L02-04-bertrand · Next: L02-06-hotelling-voting · Map: L02-00-map

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1 · U — the story (Hardin 1968)

A pasture shared by herders: each extra animal benefits its owner, but the degradation it causes is shared by all. Each herder's incentive is to grow his herd, and everyone ends up worse off: the free-rider problem, and the prisoner's dilemma with continuous strategies (Tadelis p. 84). Hardin's latter-day commons: the atmosphere, oceans, fish stocks, parking meters.

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2 · C — Tadelis's clean-air game (pp. 84–86)

n players; each chooses $k_i \geq 0$ of a common stock K of clean air; the benefit from own use is $\ln k_i$ and every player also enjoys what is left:

$$u_i(k_i, k_{-i}) = \ln k_i + \ln\Big(K - \sum_{j=1}^{n} k_j\Big) \qquad \text{(eq. 5.1)}$$

(assuming $\sum_j k_j < K$, footnote 3).

Best response. Given k_{-i},

$$\frac{\partial u_i}{\partial k_i} = \frac{1}{k_i} - \frac{1}{K - \sum_j k_j} = 0 \iff K - k_i - \sum_{j \neq i} k_j = k_i \iff \boxed{BR_i(k_{-i}) = \frac{K - \sum_{j \neq i} k_j}{2}}$$

The second derivative $-1/k_i^2 - 1/(K - \sum k_j)^2 < 0$ confirms a maximum. Slope $-1/2$ in each rival's use: strategic substitutes, like Cournot (Tadelis p. 93).

> **Method (p. 85).** When the best responses are functions, a Nash equilibrium is a solution of n equations $k_i = BR_i(k_{-i})$ in n unknowns.

Two players. $k_1 = (K - k_2)/2$ and $k_2 = (K - k_1)/2$. Substitute the second into the first: $k_1 = (K - (K - k_1)/2)/2 = K/4 + k_1/4$, so $\frac{3}{4}k_1 = K/4$ and

$$k_1^* = k_2^* = \frac{K}{3}.$$

The Pareto benchmark (p. 86). Maximize the "world's payoff" $w = \sum_i u_i = \ln k_1 + \ln k_2 + 2\ln(K - k_1 - k_2)$:

$$\frac{\partial w}{\partial k_1} = \frac{1}{k_1} - \frac{2}{K - k_1 - k_2} = 0, \qquad \frac{\partial w}{\partial k_2} = \frac{1}{k_2} - \frac{2}{K - k_1 - k_2} = 0.$$

So $k_1 = k_2 = k$ and $K - 2k = 2k$, giving

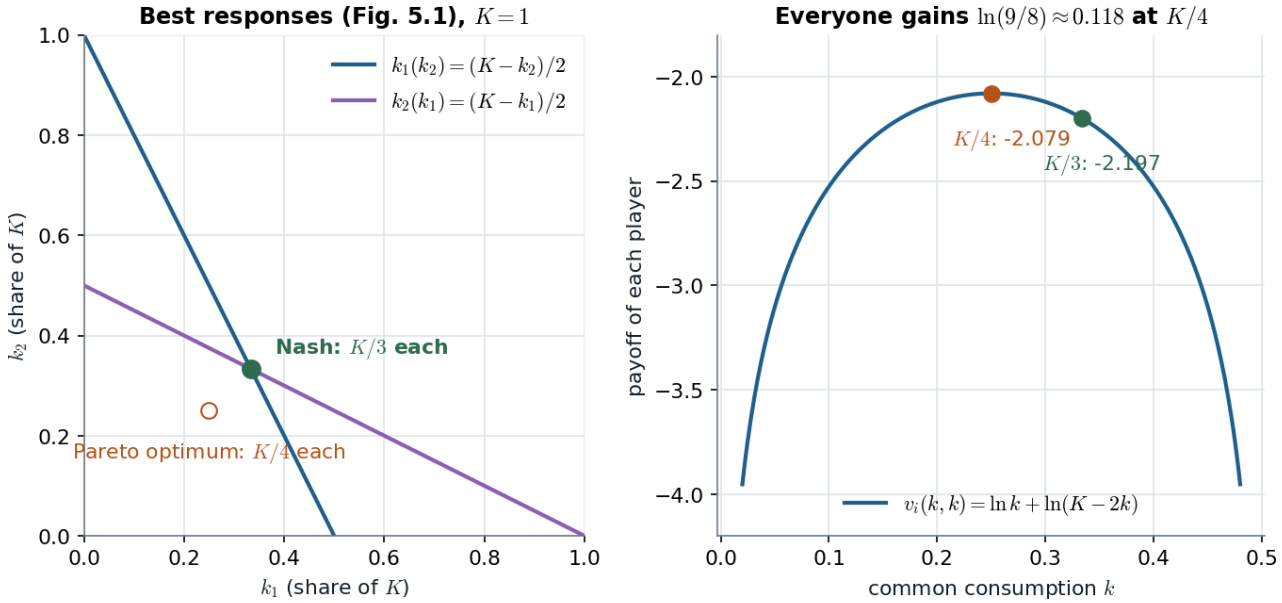
$$k_1 = k_2 = \frac{K}{4}.$$

> **The difference is a single factor of 2.** The private FOC charges the player **one** unit of the harm $(1/(K - \sum k))$; the social FOC charges **two**, because the remaining air is enjoyed by both. Each player ignores the harm to the other: an externality, so the equilibrium over-consumes.

Both gain (footnote 5, p. 87). Each player's payoff at $K/4$ minus at $K/3$:

$$\Delta u_i = \left[\ln \tfrac{K}{4} + \ln \tfrac{K}{2}\right] - \left[\ln \tfrac{K}{3} + \ln \tfrac{K}{3}\right] = 2\ln 3 - \ln 2 - \ln 4 = \ln \tfrac{9}{8} \approx 0.118 > 0.$$

The printed footnote gives 0.051; that is the same expression in **base-10** logarithms ($2\log_{10} 3 - \log_{10} 8 = 0.0512$). With the natural logarithm of eq. (5.1) the gain is 0.118. The sign, which is the point, is the same.



Left: Fig. 5.1 with $K = 1$; the equilibrium sits on both lines, the optimum on neither. Right: the symmetric payoff $\ln k + \ln(K - 2k)$ peaks at $K/4$ (-2.079); the equilibrium $K/3$ gives -2.197. Source: eq. (5.1), pp. 85–87.

The right panel is the tragedy in one curve: the equilibrium is to the right of the peak, on the falling side, so a uniform cut in everyone's use raises everyone's payoff. No single player will make that cut alone, because on his own best-response line a unilateral cut lowers his payoff.

Tadelis closes (p. 87) with Hardin's policy point and its counterweight: regulation could help everyone, if the regulator can be trusted, a question game theory alone cannot answer. The n-player version is Tadelis exercise 5.7 (p. 96).

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3 · C — Gibbons's village green (§1.2.D, pp. 27–29)

n farmers choose goats g_i; $G = \sum_i g_i$; each goat costs c and is worth $v(G)$, with $v' < 0$, $v'' < 0$ below $G_{\max}$. Payoff (1.2.4): $g_i v(g_1 + \cdots + g_n) - c g_i$.

FOC of farmer i (1.2.5): $v(g_i + g^*_{-i}) + g_i v'(g_i + g^*_{-i}) - c = 0$. The first term is the value of one more goat, the second the harm it does to **his own** g_i goats.

Sum over i and divide by n (1.2.6): $v(G^*) + \frac{1}{n} G^* v'(G^*) - c = 0$.

Social optimum (1.2.7): $\max_G Gv(G) - Gc$ gives $v(G^{**}) + G^{**} v'(G^{**}) - c = 0$.

Proof that $G^* > G^{**}$ (footnote 12, p. 28). Suppose instead $G^* \le G^{**}$. Then $v(G^*) \ge v(G^{**})$ ($v' < 0$); $0 > v'(G^*) \ge v'(G^{**})$ ($v'' < 0$); and $G^*/n < G^{**}$. So $\frac{1}{n} G^* v'(G^*) > G^{**} v'(G^{**})$ (a smaller positive number times a smaller-in-size negative number). Hence the left side of (1.2.6) **strictly exceeds** the left side of (1.2.7). Both equal zero: contradiction. ■

> **Read the two FOCs side by side:** the farmer weighs the marginal harm by **his own herd**; society weighs it by **the whole herd**. Too many goats graze.

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4 · U — where this goes in an organization

Any shared input with private use and common cost follows the same arithmetic: a shared IT budget, a common sales pool, team effort with a shared output (Lecture 1's partnership game was its discrete cousin). Lecture 5 asks whether repetition can sustain the cooperative $K/4$ (Lecture 5).

Links

- Neighbours: L02-03-cournot (same strategic-substitutes structure), L02-02-nash-equilibrium §5 (NE need not be Pareto optimal).
- Lecture 1 (Pareto, social dilemmas): Lecture 1.
- Code: L02-12-python §5.
- Practice: Tadelis 5.7, 5.9 (pp. 96–97); Muñoz-Garcia & Toro-Gonzalez Ch. 2 (pp. 37–84).

5.2.5 · Political ideology and electoral competition: both candidates go to the middle

Claim. Two office-seeking candidates choose platforms on a left–right line of 101 citizens, each citizen voting for the nearer platform. Against a rival at $a_j > 0$ any platform strictly between $-a_j$ and a_j wins, so the best response is an **interval**, and the intervals of the two players meet in a single point: **both at the median citizen,** $a_1 = a_2 = 0$. This is Hotelling's (1929) insight and the seed of the median voter theorem, and it is Lecture 1's pharmacy case solved with best responses instead of dominance.

Not covered in class. The Lecture 2 slides (`Aula/Lecture02.pdf`) do not teach Hotelling voting (slide 3 only recalls Lecture 1's pharmacies and says that with prices the chains would move apart, a problem-set question). This note is kept as reading and practice; it is examinable only if a problem set brings it.

Prev: L02-05-commons · Next: L02-07-mixed-strategies · Map: L02-00-map

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1 · C — the game (pp. 93–94)

- Players: two candidates who care **only about winning**: win $\succ$ tie $\succ$ lose.
- Citizens: 101, labelled $-50, \dots, 0, \dots, +50$ by ideology; each votes for the nearer platform, and an indifferent citizen tosses a coin (or abstains, footnote 9).
- Strategies: $a_i \in \{-50, \dots, +50\}$. Majority wins.

Worked (Fig. 5.4, p. 94). $a_1 = -15$, $a_2 = +22$. Citizen 3 is at distance 18 from -15 and 19 from $+22$, so votes for candidate 1; citizen 4 is at 19 and 18, so votes for candidate 2. Candidate 1 gets $-50, \dots, 3$: **54 votes**; candidate 2 gets $4, \dots, 50$: **47 votes**. Candidate 1 wins: being nearer the middle wins.

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2 · C — the best-response correspondence (p. 94)

Fix $a_j > 0$. Candidate i at a_i:

a_i	Outcome	Why
$a_i = a_j$ or $a_i = -a_j$	tie	identical platforms split everything; mirror platforms split the line at 0
$a_i > a_j$ or $a_i < -a_j$	lose	the rival is nearer to more than half the citizens
$-a_j < a_i < a_j$	**win**	the midpoint $(a_i + a_j)/2$ lies right of 0, so i gets every citizen ≤ 0 plus some more

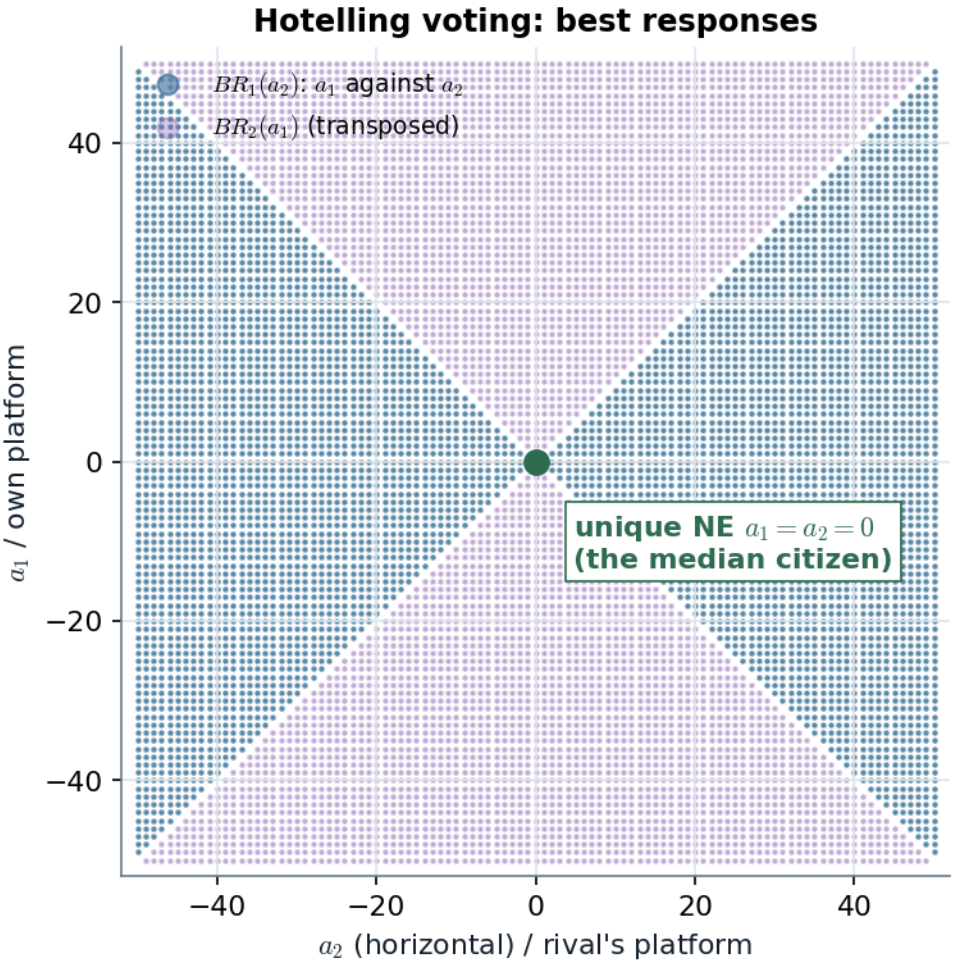
So for integers

$$BR_i(a_j) = \begin{cases} [a_j + 1,\ -a_j - 1] & a_j < 0 \\ \{0\} & a_j = 0 \\ [-a_j + 1,\ a_j - 1] & a_j > 0 \end{cases}$$

Against $a_j = 0$ every other platform loses and 0 ties, so the best response is 0. **Against** $a_j = \pm 1$ the interval collapses to $\{0\}$ (footnote 10).

Equilibrium. Suppose $a_j \neq 0$. Then $BR_i(a_j)$ lies strictly inside $(-|a_j|, |a_j|)$, so $|a_i| < |a_j|$; but the same argument for j requires $|a_j| < |a_i|$. Contradiction. Hence $a_j = 0$, and then $BR_i(0) = \{0\}$:

$a_1^* = a_2^* = 0$ is the unique Nash equilibrium



Each column of dots is $BR_1(a_2)$; the transposed set is $BR_2(a_1)$. The two bow-ties overlap only at the centre, the unique equilibrium. Computed by brute force over all 101^2 platform pairs. Source: Tadelis §5.2.5, pp. 93–95.

The bow-tie shape is the content: the further the rival moves from the centre, the **more** platforms beat him, but all of them are closer to the centre than he is, so nobody can rest anywhere but the middle.

Hotelling (1929, p. 54), quoted by Tadelis (p. 94): the parties do not offer "two strongly contrasted positions"; "each party strives to make its platform as much like the other's as possible."

3 · C — the median voter theorem and the continuum (pp. 94–95)

Median voter theorem (Black 1948; Downs 1957): with voters ordered on one dimension and **single-peaked** preferences (footnote 11: utility highest at the bliss point and falling on both sides), the platform at the median voter beats any other in a simple majority vote.

Remark (p. 95). With a continuum of voters on $[\underline{a}, \overline{a}]$ with distribution F, the median a_m solves $\Pr\{a \leq a_m\} = \frac{1}{2}$ ($a_m = (\underline{a} + \overline{a})/2$ for uniform F). For $a_j \neq a_m$ the best response is an **open** interval: between a_j and the "opposite" platform a' that ties with it. Only $a_j = a_m$ has a unique best reply, a_m itself, so both at a_m is the unique NE (Tadelis exercise 5.15, p. 99).

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4 · U — the link to Lecture 1's pharmacies

Lecture 1's mini case (two chains on a 7-block avenue) is the **same model with customers instead of voters and a payoff equal to the number of customers instead of win/tie/lose**. There IESDS removed the outer blocks round by round until both chains stood in the middle; here best responses deliver the centre directly. Both predict **clustering**, which is excessive from the walking-distance point of view: total distance falls when the two stores spread out toward the two halves of the avenue. The prediction depends on fixed prices and one dimension; Tadelis exercise 5.16 (p. 99) adds price competition.

Links

- Neighbours: L02-02-nash-equilibrium, L02-01-best-response-rationalizability (best-response correspondences).
- Lecture 1 (the pharmacy case by IESDS): Lecture 1. Cabral's linear city (Ch. 14) is the IO version.
- Code: L02-12-python §6.
- Practice: Tadelis 5.15 (p. 99), 4.10 (p. 78, the popsicle stands by rationalizability); Gibbons 1.8 (p. 50).

Slides 42–55 · 6.1–6.2 · Mixed strategies: each player's mix is pinned by the *other* player's indifference

Claim. A mixed strategy is a probability distribution over pure strategies; a belief is a distribution over the rivals' strategies. A mixed-strategy NE is a profile of mutual best responses, and **a player mixes only between strategies that pay the same** (Prop. 6.1). So the recipe is: find the rival's mix that makes **me** indifferent; that is the rival's equilibrium strategy. The lecture's example is the Battle of the Sexes, whose two pure equilibria gain a third, mixed one, $(p, q) = (\frac{2}{3}, \frac{1}{3})$, worth only $\frac{2}{3}$ to each player (slides 47–50). The book's other examples (Matching Pennies, rock-paper-scissors, the §6.2.3 game) are reading.

Prev: L02-06-hotelling-voting · Next: L02-08-iesds-rationalizability-revisited · Map: L02-00-map

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1 · U — why mix (slide 42), and games with no pure equilibrium (reading)

Slide 42's motivation: **a player randomizes to avoid being predictable**, to surprise or to avoid being surprised. The book's examples below (Matching Pennies, rock-paper-scissors) were not covered in class.

Matching Pennies (p. 101). Player 1 wins if the pennies match:

1 2	H	T
H	1, −1	−1, 1
T	−1, 1	1, −1

Underline method: player 1 wants to match, player 2 to mismatch, so **no cell is a mutual best reply**. The same holds in rock-paper-scissors (p. 102): from any pure pair at least one player switches. Gibbons (pp. 29–30): poker (bluffing), pitcher vs batter, attack by land or sea: games where each wants to **outguess** the other. Tadelis (p. 111): penalty kicks, bosses monitoring employees, police and criminals.

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2 · C — the definitions, exactly

Definition 6.1 (p. 102; slide 42). A **mixed strategy** of player i is a probability distribution over S_i: $\sigma_i : S_i \to [0, 1]$ with $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$. The set of mixed strategies is $\Sigma_i := \Delta(S_i)$ (Tadelis writes ΔS_i, the simplex), and a mixed-strategy profile is $\sigma = (\sigma_1, \dots, \sigma_n) \in \Sigma := \Sigma_1 \times \cdots \times \Sigma_n$. Every pure strategy is a (degenerate) mixed strategy, and with mixed strategies players evaluate **expected** payoffs (slide 42).

Definition 6.2 (p. 104). s_i is in the **support** of σ_i iff $\sigma_i(s_i) > 0$.

Definition 6.3 (p. 104). If S_i is an interval, a mixed strategy is a CDF $F_i(x) = \Pr\{s_i \le x\}$; with density f_i, the support is $\{s_i : f_i(s_i) > 0\}$. Tadelis's example: in Cournot with capacity 100, choose q_i uniformly on $[30, 50]$: $F_i(s) = (s - 30)/20$ on $[30, 50]$, $f_i = 1/20$ there (Fig. 6.1).

Definition 6.4 (p. 105). A **belief** of player i is $\pi_i \in \Delta S_{-i}$, with $\pi_i(s_{-i})$ the probability i assigns to s_{-i}. Beliefs live in the same set as the rivals' mixed profiles.

Definition 6.5 (p. 105; slide 43). Expected payoffs, in the slide's three formulas (players randomize independently, so the probability of a pure profile is the product of the individual probabilities):

$$u_i(\sigma) = \sum_{s \in S} \sigma_1(s_1) \cdots \sigma_n(s_n)\, u_i(s), \tag{1}$$

$$u_i(s_i, \sigma_{-i}) = \sum_{s_{-i} \in S_{-i}} \Big(\prod_{j \neq i} \sigma_j(s_j) \Big) u_i(s_i, s_{-i}), \tag{2}$$

$$u_i(\sigma_i, \sigma_{-i}) = \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(s_i, \sigma_{-i}). \tag{3}$$

> **A mixed strategy's payoff is a weighted average of its pure strategies' payoffs.** Everything below follows from that one line: no mix can beat the best pure strategy in its support, and a mix ties with it only if every pure strategy in the support ties.

Worked (p. 106; reading). RPS against $\sigma_2 = (\frac{1}{2}, \frac{1}{2}, 0)$: $u_1(R, \sigma_2) = \frac{1}{2} \cdot 0 + \frac{1}{2}(-1) + 0 \cdot 1 = -\frac{1}{2}$; $u_1(P, \sigma_2) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 0 + 0 \cdot (-1) = \frac{1}{2}$; $u_1(S, \sigma_2) = \frac{1}{2}(-1) + \frac{1}{2} \cdot 1 + 0 = 0$. Unique best reply: **P**.

Worked, continuous (pp. 106–107; reading): the all-pay auction for a dollar. Both bid, both pay, the higher bid wins. Against $s_2 \sim U[0,1]$, a bid $s_1 < 1$ gives

$$u_1 = \Pr\{s_1 < s_2\}(-s_1) + \Pr\{s_1 > s_2\}(1 - s_1) = (1 - s_1)(-s_1) + s_1(1 - s_1) = 0,$$

(ties have probability 0), and any bid above 1 gives $1 - s_1 < 0$. **Every bid in** $[0,1]$ **earns exactly 0**: player 1 is indifferent across a continuum. (Uniform bidding is in fact a symmetric equilibrium: Tadelis exercise 6.10, p. 125.)

Definition 6.6 (p. 107; slide 44). $\sigma^* \in \Sigma$ is a **Nash equilibrium** if for every $i \in I$, σ_i^* is a best response to σ_{-i}^*:

$$u_i(\sigma_i^*, \sigma_{-i}^*) \;\geq\; u_i(\sigma_i, \sigma_{-i}^*) \qquad \forall \sigma_i \in \Sigma_i. \tag{4}$$

Reading σ_{-i}^* as i's belief: rational play given beliefs (rationalizability with uncertain beliefs), plus correct beliefs (p. 108).

Proposition 6.1 (p. 108). If σ^* is a NE and s_i, s_i' are both in the support of σ_i^*, then

$$u_i(s_i, \sigma_{-i}^*) = u_i(s_i', \sigma_{-i}^*) = u_i(\sigma_i^*, \sigma_{-i}^*).$$

Proof. Suppose $u_i(s_i, \sigma_{-i}^*) > u_i(s_i', \sigma_{-i}^*)$. Move the probability $\sigma_i^*(s_i') > 0$ from s_i' to s_i. By Def. 6.5 the payoff rises by $\sigma_i^*(s_i')\,[u_i(s_i, \sigma_{-i}^*) - u_i(s_i', \sigma_{-i}^*)] > 0$, so σ_i^* was not a best response. Hence the two are equal, and the weighted average of equal numbers equals each of them. ∎

Tadelis's Prop. 6.1 is one direction of one condition. The lecture states and proves the **full characterisation**:

2b · C — the principle of indifference (slides 44–46)

Support. $\operatorname{supp} \sigma_i := \{s_i \in S_i : \sigma_i(s_i) > 0\}$ (Tadelis Def. 6.2).

Principle of indifference (slide 44). $\sigma^* \in \Sigma$ is a Nash equilibrium **if and only if**, for every $i \in I$,

$$\text{(i)} \quad u_i(\sigma^*) = u_i(s_i, \sigma_{-i}^*) \quad \text{for every } s_i \in \operatorname{supp} \sigma_i^*; \tag{5}$$

$$\text{(ii)} \quad u_i(\sigma^*) \geq u_i(s_i, \sigma_{-i}^*) \quad \text{for every } s_i \in S_i \setminus \operatorname{supp} \sigma_i^*. \tag{6}$$

In words: **every strategy played with positive probability yields the same expected payoff, and no strategy outside the support yields more.** This is MWG Prop. 8.D.1 (p. 250).

Proof, indifference ⇒ Nash (slide 45). Suppose σ^* satisfies (5) and (6) for every i. Together they say $u_i(s_i, \sigma^*_{-i}) \le u_i(\sigma^*_i, \sigma^*_{-i})$ for **every** pure $s_i \in S_i$. Take any $\sigma_i \in \Sigma_i$; by (3),

$$\begin{aligned} u_i(\sigma_i, \sigma^*_{-i}) &= \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(s_i, \sigma^*_{-i}) \\ &\le \sum_{s_i \in S_i} \sigma_i(s_i)\, u_i(\sigma^*_i, \sigma^*_{-i}) \\ &= u_i(\sigma^*_i, \sigma^*_{-i}), \end{aligned}$$

because the weights $\sigma_i(s_i)$ are non-negative and sum to 1. That is (4). ■

Proof, Nash ⇒ indifference (slide 46). Suppose σ^* is a NE but (5) fails for some player j: there are $s'_j, s''_j \in \operatorname{supp} \sigma^*_j$ with $u_j(s''_j, \sigma^*_{-j}) > u_j(s'_j, \sigma^*_{-j})$. Build $\hat{\sigma}_j$ by moving the probability of s'_j onto s''_j:

$$\hat{\sigma}_j(s_j) = \begin{cases} \sigma^*_j(s_j) & s_j \notin \{s'_j, s''_j\}, \\ 0 & s_j = s'_j, \\ \sigma^*_j(s'_j) + \sigma^*_j(s''_j) & s_j = s''_j. \end{cases}$$

By (3) the payoff changes by $\sigma^*_j(s'_j)\,[u_j(s''_j, \sigma^*_{-j}) - u_j(s'_j, \sigma^*_{-j})] > 0$, so $u_j(\hat{\sigma}_j, \sigma^*_{-j}) > u_j(\sigma^*_j, \sigma^*_{-j})$, contradicting (4). Condition (6) follows the same way: if some $s_j \notin \operatorname{supp} \sigma^*_j$ paid more than $u_j(\sigma^*)$, moving all the weight onto s_j would pay more. ■

> **How to use it.** To find a mixed NE, guess the supports, then write the indifference conditions (5): player i's indifference is an equation in the **others'** probabilities. Then check (6) for the unused strategies. To verify a candidate, it suffices to check deviations to **pure** strategies.

2c · Worked — Example 3, the Battle of the Sexes in mixed strategies (slides 47–50)

Notation. $p = \Pr(\text{Alex plays O}) \in [0,1]$, $q = \Pr(\text{Chris plays O}) \in [0,1]$; payoffs as in L02-01-best-response-rationalizability §1 (Alex 2 and Chris 1 at (O, O); Alex 1 and Chris 2 at (F, F); 0 apart).

Alex's expected payoffs (slide 47), from (2):

$$u_A(O, q) = 2q + 0\,(1-q) = 2q, \tag{7}$$

$$u_A(F, q) = 0\,q + 1\,(1-q) = 1-q. \tag{8}$$

Alex prefers O iff (7) > (8): $2q > 1-q$, i.e. $q > \frac{1}{3}$. So (slide 48)

$$BR_A(q) = \begin{cases} F\ (p=0) & q < \frac{1}{3}, \\ \text{any } p \in [0,1] & q = \frac{1}{3}, \\ O\ (p=1) & q > \frac{1}{3}. \end{cases} \tag{9}$$

Chris's expected payoffs (slide 49):

$$u_C(O, p) = 1 \cdot p + 0\,(1-p) = p, \tag{10}$$

$$u_C(F, p) = 0 \cdot p + 2\,(1-p) = 2 - 2p. \tag{11}$$

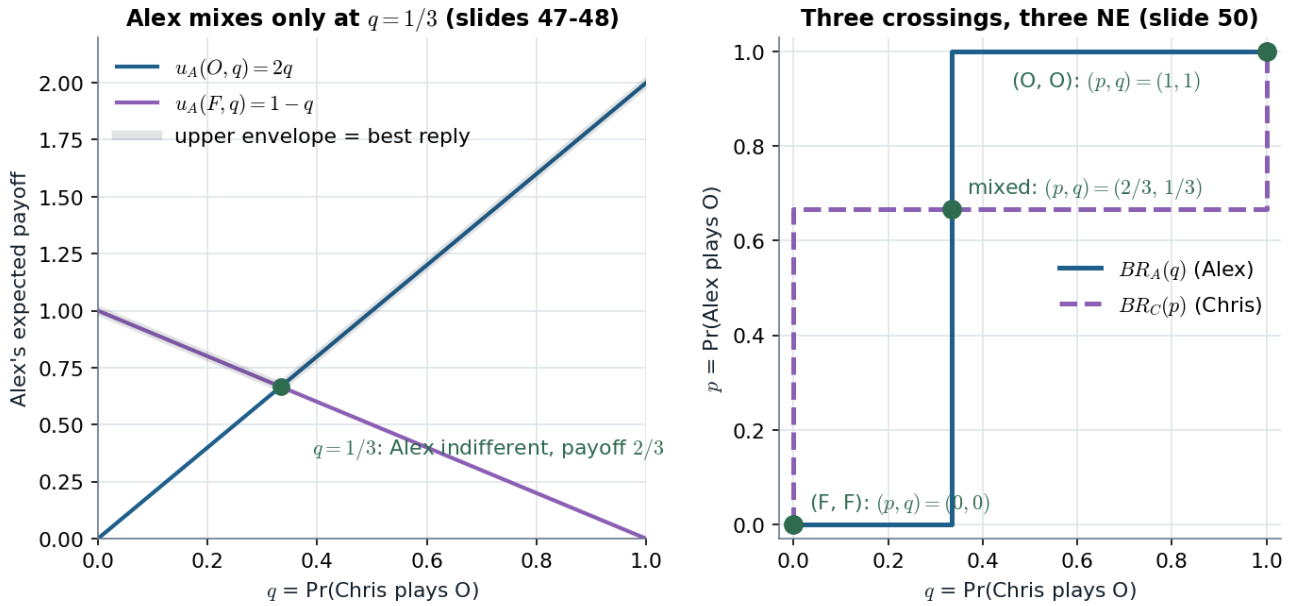
Chris prefers O iff (10) > (11): $p > 2 - 2p$, i.e. $p > \frac{2}{3}$. So

$$BR_C(p) = \begin{cases} q = 0 \ (F) & p < \frac{2}{3}, \\ q \in [0,1] & p = \frac{2}{3}, \\ q = 1 \ (O) & p > \frac{2}{3}. \end{cases} \tag{12}$$

All Nash equilibria (slide 50) are the intersections of (9) and (12):

(p, q)	Profile	Alex's payoff	Chris's payoff
$(0, 0)$	(F, F)	1	2
$(1, 1)$	(O, O)	2	1
$(\frac{2}{3}, \frac{1}{3})$	**mixed**	$2q = 2 \cdot \frac{1}{3} = \frac{2}{3}$	$p = \frac{2}{3}$

At the mixed equilibrium both supports are full, so (5) is the whole check: (7) = (8) at $q = \frac{1}{3}$ and (10) = (11) at $p = \frac{2}{3}$.



Left: eqs. (7)–(8); the upper envelope is Alex's best-reply payoff, and Alex is willing to mix only at $q = \frac{1}{3}$. Right: eqs. (9) and (12), three crossings, three equilibria. Source: slides 47–50.

Two things to read off. **Each probability is pinned down by the other player's payoffs**: Alex's $p = \frac{2}{3}$ is what makes **Chris** indifferent in (10)–(11), and Chris's $q = \frac{1}{3}$ is what makes **Alex** indifferent in (7)–(8). And **each player expects $\frac{2}{3}$, less than in either pure equilibrium** (where the worse-off player still gets 1), because they miscoordinate with probability $p(1-q) + (1-p)q = \frac{2}{3} \cdot \frac{2}{3} + \frac{1}{3} \cdot \frac{1}{3} = \frac{5}{9}$.

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3 · Not covered in class — Matching Pennies (§6.2.1, pp. 108–111)

Let $p = \Pr(1 \text{ plays H})$, $q = \Pr(2 \text{ plays H})$.

Player 1's payoffs (6.1)–(6.2): $u_1(H, q) = q - (1-q) = 2q - 1$; $u_1(T, q) = -q + (1-q) = 1 - 2q$. H is better iff $2q - 1 > 1 - 2q$ iff $q > \frac{1}{2}$. So

$$BR_1(q) = \begin{cases} p = 0 & q < \frac{1}{2} \\ p \in [0,1] & q = \frac{1}{2} \\ p = 1 & q > \frac{1}{2} \end{cases} \qquad BR_2(p) = \begin{cases} q = 1 & p < \frac{1}{2} \\ q \in [0,1] & p = \frac{1}{2} \\ q = 0 & p > \frac{1}{2} \end{cases}$$

(player 2: $u_2(p, H) = -p + (1-p) = 1 - 2p$, $u_2(p, T) = 2p - 1$). The correspondences cross once:

$$\boxed{(p, q) = (\tfrac{1}{2}, \tfrac{1}{2})}$$

> **The indifference logic (p. 111):** player 1 is willing to mix only if $q = \frac{1}{2}$, so **player 1's indifference pins player 2's strategy**, and vice versa. Each player mixes in a way that makes the *other* indifferent. A player's own payoffs do not determine his own mix.

Gibbons writes the same payoff as one linear function of the mixing probability, $(2q-1) + r(2-4q)$ (eq. 1.3.1, p. 34; his r is the probability of Heads for player 1): increasing in r when $q < \frac{1}{2}$ in his orientation, flat at $q = \frac{1}{2}$. Flat means **every** r is a best reply: that is why the correspondence has a vertical segment.

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4 · Not covered in class — rock-paper-scissors (§6.2.2, pp. 111–113)

Claim 6.1. No equilibrium has one player pure and the other mixing: against a pure strategy the other player's three payoffs are all different (0, 1, −1 in some order), so he cannot be indifferent, so he plays pure; and there is no pure equilibrium.

Claim 6.2. No equilibrium has a player mixing over exactly two strategies. If i mixes R and P, then for j, P beats R against both (j's payoffs: vs R, P gives $1 > 0$; vs P, P gives $0 > -1$), so j never plays R; but if j never plays R, then for i, S beats P against both remaining (vs P: $1 > 0$; vs S: $0 > -1$), so i never plays P, contradicting the mix. Same for every pair.

Full support. Let i play $(\sigma_R, \sigma_P, 1 - \sigma_R - \sigma_P)$. Player j's payoffs:

$$u_j(R) = -\sigma_P + (1 - \sigma_R - \sigma_P) = 1 - \sigma_R - 2\sigma_P, \quad u_j(P) = \sigma_R - (1 - \sigma_R - \sigma_P) = 2\sigma_R + \sigma_P - 1, \quad u_j(S) = -\sigma_R + \sigma_P.$$

$u_j(R) = u_j(S)$: $1 - \sigma_R - 2\sigma_P = -\sigma_R + \sigma_P \Rightarrow \sigma_P = \frac{1}{3}$. $u_j(P) = u_j(S)$: $2\sigma_R + \sigma_P - 1 = -\sigma_R + \sigma_P \Rightarrow \sigma_R = \frac{1}{3}$. Hence $\sigma_S = \frac{1}{3}$, and

$$\sigma_1^* = \sigma_2^* = (\tfrac{1}{3}, \tfrac{1}{3}, \tfrac{1}{3}) \text{ is the unique NE, each player's payoff 0.}$$

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5 · Not covered in class — pure and mixed together (§6.2.3, pp. 113–114)

1 2	C	R
M	0, 0	3, 5
D	4, 4	0, 3

Pure NE (underline method): against C player 1 prefers D ($4 > 0$); against R, M ($3 > 0$); against M player 2 prefers R ($5 > 0$); against D, C ($4 > 3$). Mutual: **(M, R)** and **(D, C)**.

Mixed: $p = \sigma_1(M)$, $q = \sigma_2(C)$.

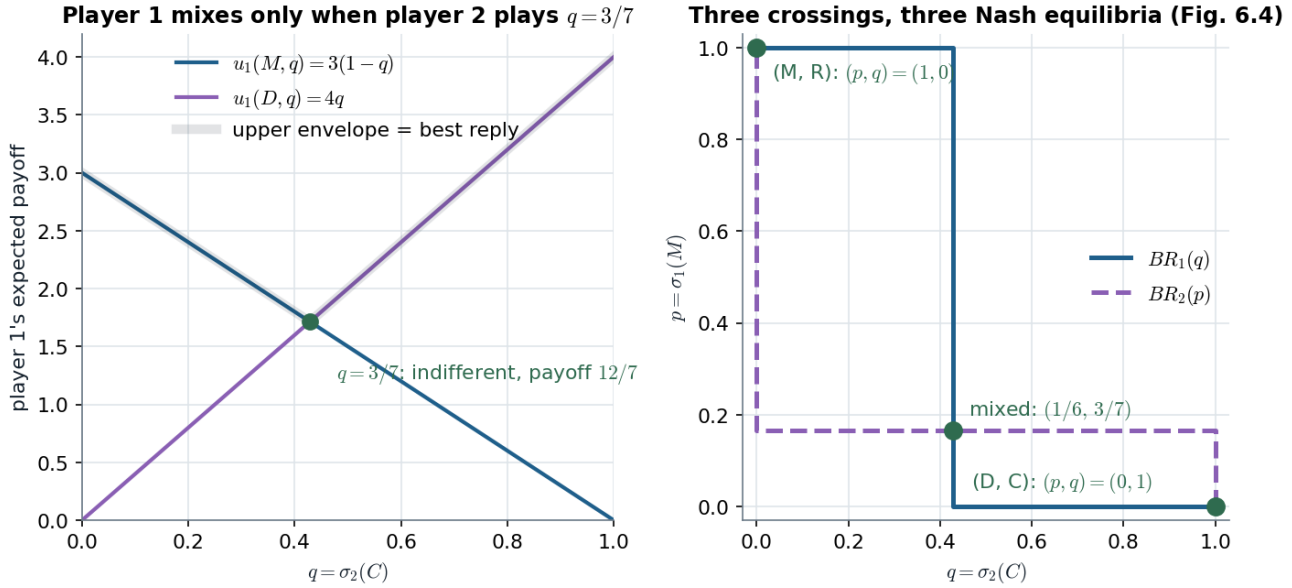
- Player 1 indifferent: $u_1(M,q) = 3(1-q)$, $u_1(D,q) = 4q$; $3 - 3q = 4q \Rightarrow \boxed{q = \frac{3}{7}}$.
- Player 2 indifferent: $u_2(p,C) = 4(1-p)$, $u_2(p,R) = 5p+3(1-p) = 3+2p$; $4-4p = 3+2p \Rightarrow \boxed{p = \frac{1}{6}}$.

$$\sigma^* = ((\tfrac{1}{6}, \tfrac{5}{6}), (\tfrac{3}{7}, \tfrac{4}{7})).$$

Payoffs there: player 1 gets $4q = \frac{12}{7}$; player 2 gets $3 + 2p = \frac{10}{3}$. Best responses:

$$BR_1(q) = \begin{cases} p = 1 & q < \frac{3}{7} \\ [0,1] & q = \frac{3}{7} \\ p = 0 & q > \frac{3}{7} \end{cases} \qquad BR_2(p) = \begin{cases} q = 1 & p < \frac{1}{6} \\ [0,1] & p = \frac{1}{6} \\ q = 0 & p > \frac{1}{6} \end{cases}$$

They cross three times: $(p,q) \in \{(1,0), (\frac{1}{6}, \frac{3}{7}), (0,1)\}$.



Left: the analogue of Tadelis Fig. 6.2 for the §6.2.3 game; the upper envelope is player 1's best-reply payoff. Right: Fig. 6.4, three crossings, three equilibria. Source: pp. 113–114.

Read the left panel to see why player 1's mix is irrelevant to player 1: at $q = \frac{3}{7}$ both lines give $\frac{12}{7}$, so every p is optimal; what fixes $p = \frac{1}{6}$ is player 2's indifference, on the other side of the game.

> **"Almost always" an odd number (footnote 6, p. 113).** A 2×2 game with two pure equilibria generically has a third, mixed one; generic finite games have an odd number of equilibria (the index theorem, named only).

> **Interactive companion.** The Mixing Square (`companions/companion-mixed-2x2.html`) — the default is this game; set the belief sliders to $q = \frac{3}{7}$, $p = \frac{1}{6}$ with "Beliefs to indifference" and watch both gain read-outs hit zero; then load Matching Pennies, the Battle of the Sexes and the Stag Hunt.

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6 · Not covered in class — every 2×2 game, by cases (Gibbons §1.3.B, pp. 41–44)

Write player 1's payoffs as $x = u_1(\text{Up}, \text{Left})$, $y = u_1(\text{Up}, \text{Right})$, $z = u_1(\text{Down}, \text{Left})$, $w = u_1(\text{Down}, \text{Right})$ (Fig. 1.3.8), $q = \Pr(\text{Left})$.

Case	Condition	$r^*(q)$, with $r = \Pr(\text{Up})$
(i)	$x > z$, $y > w$	Up strictly dominant: $r = 1$
(ii)	$x < z$, $y < w$	Down strictly dominant: $r = 0$
(iii)	$x > z$, $y < w$	Up iff $q > q'$, any r at $q = q'$
(iv)	$x < z$, $y > w$	Up iff $q < q'$, any r at $q = q'$

with $q' = \dfrac{w - y}{(x - z) + (w - y)}$. Equalities ($x = z$ or $y = w$) give L-shaped correspondences. Pairing any of player 1's four shapes with any of player 2's always produces **at least one crossing**, and the generic outcomes are (1) one pure NE, (2) one mixed NE (Matching Pennies), or (3) two pure and one mixed (Battle of the Sexes, Tadelis §6.2.3). This is the graphical existence proof for 2×2 games, previewing L02-09-existence.

Battle of the Sexes (Gibbons p. 40, Tadelis exercise 6.1, p. 123). The lecture's Example 3, worked in §2c: **each goes to his own favourite event with probability** $\frac{2}{3}$, and they miscoordinate with probability $\frac{5}{9}$. In Gibbons's case language it is case (iii) for both players: two pure equilibria and one mixed.

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7 · C — what a mixed strategy means, and existence (slide 51)

The slides list three **common interpretations** of mixed strategies:

Interpretation	What σ_j is
Beliefs	σ_{-i} describes what player i **believes** the others will do
Empirical frequencies	long-run frequencies of actions in **repeated** interactions
Population distribution	the **share of a population** of players using each action

Theorem (Nash, 1950; slide 51). Every finite normal-form game has a Nash equilibrium in mixed strategies (L02-09-existence). Slide 51's "why?": a profile of pure strategies is also a profile of mixed strategies, because each pure s_i is the degenerate σ_i with $\sigma_i(s_i) = 1$; so a pure equilibrium, when it exists, is one of the equilibria the theorem counts (the Battle of the Sexes has three).

The book's version of the beliefs reading:

Gibbons (pp. 39–40), following Harsanyi (1973): a mixed equilibrium **does not require anyone to flip a coin**. Player j's mix represents player i's **uncertainty** about j's pure choice, e.g. a pitcher who picks the pitch from private information the batter does not see. Tadelis (p. 105) offers the same reading: an opponent whose unknown background determines his play. rules/game-theory item 8: the indifference condition **pins the opponent's mix**, which is the useful part. Tadelis returns to this "purification" in §12.5 (Lecture 7).

8 · G → C — MWG Prop. 8.D.1 (p. 250) is the slides' principle of indifference

Promoted to the course layer by the lecture (§2b). MWG Def. 8.D.2 (p. 250) is Def. 6.6. **Prop. 8.D.1 (p. 250):** with S_i^+ the support of σ_i, σ is a NE iff for every i (i) $u_i(s_i, \sigma_{-i}) = u_i(s_i', \sigma_{-i})$ for all $s_i, s_i' \in S_i^+$, and (ii) $u_i(s_i, \sigma_{-i}) \geq u_i(s_i', \sigma_{-i})$ for all $s_i \in S_i^+$, $s_i' \notin S_i^+$. Condition (ii) is the half Prop. 6.1 does not state: **unused strategies must not do better**. Its corollary: to test a profile it suffices to check deviations to **pure** strategies. This is the basis of support enumeration (L02-12-python §7).

9 · C — the mini case: monitoring remote work (slides 53–55; graded, method only)

Motivation (slide 53). In September 2025 Itaú dismissed employees after finding mismatches between their clock-in records and the activity monitored on their computers while working from home; the bank workers' union estimated about 1,000 dismissals (source on the slide: g1, 2025). Monitoring software is costly, and employees know it may be in place. *How much should a firm monitor, and how much shirking should it expect?*

The game (slide 54). A remote employee chooses **work** (W) or **shirk** (S); simultaneously the manager chooses **monitor** (M) or **not** (N). A shirker who is caught loses the wage.

Employee \ Manager	M	N
W	$w-g,\ v-w-h$	$w-g,\ v-w$
S	$0,\ -h$	$w,\ -w$

Parameters: $w = 10$ wage, $g = 2$ cost of effort, $v = 20$ value of output, $h = 1$ cost of monitoring.

> **Graded in-class exercise (in pairs, handed in on the day).** These notes describe the **method only**; they do not solve parts (a)–(e). Work it yourself before looking at any solution.

Method, part by part (slide 55). - *(a) No pure equilibrium.* Apply the underline method of L02-02-nash-equilibrium §3 to the four cells with the numbers plugged in: for each cell, ask whether one player gains by switching. A cell where nobody gains would be a pure NE; show that every cell has a profitable unilateral deviation. - *(b) The mixed equilibrium.* Name the probabilities (e.g. $s = \Pr(S)$ for the employee, $m = \Pr(M)$ for the manager). By the principle of indifference (§2b): the **manager's** indifference between M and N is one equation in the **employee's** shirking probability; the **employee's** indifference between W and S is one equation in the **manager's** monitoring probability. Solve each linear equation and check both lie in $(0, 1)$. - *(c)–(d) Comparative statics.* Re-solve with the new parameter ($h = 0.5$; then $w = 20$) and compare. For the intuition, ask **whose** indifference condition contains the changed parameter, and therefore **whose** probability moves (slide 50's lesson). - *(e) General formulas.* Redo (b) with w, g, h (and v) as symbols; read each probability's dependence off its formula. For the policy question, compare what each lever costs the firm; for the critique, list what the 2×2 model assumes (simultaneous moves, perfect detection when monitoring, a fixed wage, one period, no other sanctions) and ask which of these the Itaú story strains.

Related book exercises, kept out of the practice list until the case is returned: Tadelis 6.4 "Monitoring" (pp. 123–124) and 6.12 "The Tax Man" (p. 126).

Links

- Neighbours: L02-08-iesds-rationalizability-revisited (mixed strategies sharpen IESDS), L02-09-existence (why a mixed equilibrium always exists).
- Lecture 7 (Bayesian games, purification): Lecture 7.
- Code: L02-12-python §7.
- Practice: Tadelis 6.1, 6.4–6.8 (pp. 123–125); Gibbons 1.9–1.11 (pp. 50–51), 1.3 (p. 49); MWG 8.D.3–8.D.5 (p. 263); Muñoz-Garcia & Toro-Gonzalez Ch. 3 (pp. 85–151).

6.3 · IESDS and rationalizability revisited: let mixtures dominate, let beliefs be mixed

Claim. Once mixed strategies exist, a pure strategy can be **strictly dominated by a mixture** without being dominated by any pure strategy, and can be a **best response to a mixed belief** without being a best response to any pure one. Both sharpen the Lecture 1 tools: in Tadelis's 3×3 game nothing is pure-dominated, yet mixed IESDS reduces it to the 2×2 game of §6.2.3. **In two-player games, "strictly dominated" and "never a best response" are the same thing** (Prop. 6.2), so IESDS and rationalizability coincide.

Not covered in class. The Lecture 2 slides (`Aula/Lecture02.pdf`) do not teach dominance by mixed strategies or Prop. 6.2; slide 11 defines rationalizability with pure beliefs only. This note is kept as reading and practice; it is examinable only if a problem set brings it.

Prev: L02-07-mixed-strategies · Next: L02-09-existence · Map: L02-00-map

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1 · C — the two definitions (p. 114)

Definition 6.7. s_i is **strictly dominated by** $\sigma_i \in \Delta S_i$ if

$$u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i}) \qquad \forall s_{-i} \in S_{-i}.$$

Definition 6.8. σ_i is **never a best response** if there is no belief $\sigma_{-i} \in \Delta S_{-i}$ with $\sigma_i \in BR_i(\sigma_{-i})$.

Checking against pure s_{-i} is enough. The payoff against a mixed σ_{-i} is an average of payoffs against pure s_{-i} (Def. 6.5), so a strict inequality at every pure s_{-i} survives every average.

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2 · Worked — Gibbons's two pictures (pp. 31–33)

Player 1 has T, M, B; player 2 has L, R; $q = \Pr(L)$; only player 1's payoffs matter.

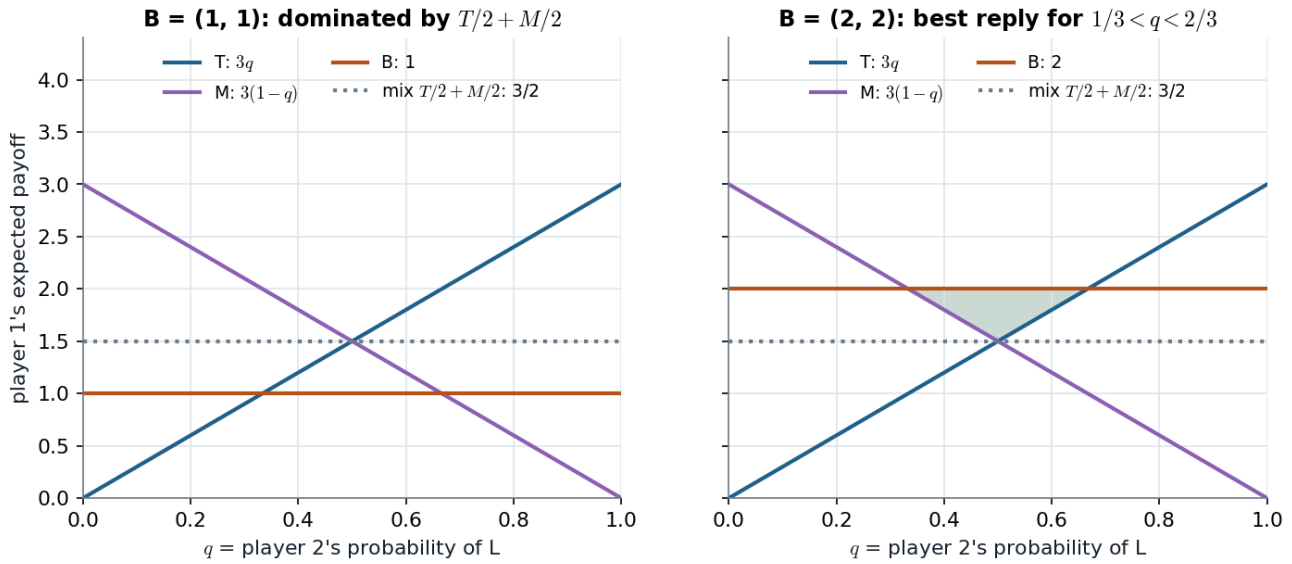
	L	R	payoff vs q
T	3	0	$3q$
M	0	3	$3(1-q)$
B (Fig. 1.3.1)	1	1	1
B (Fig. 1.3.2)	2	2	2

Fig. 1.3.1: dominated by a mix, not by a pure strategy. B is not beaten by T against R ($1 > 0$), nor by M against L. But $\frac{1}{2}T + \frac{1}{2}M$ pays $\frac{3}{2}$ against L and $\frac{3}{2}$ against R, and $\frac{3}{2} > 1$: **B is strictly dominated by the mixture**. Equivalently, the best reply is T for $q \geq \frac{1}{2}$ and M for $q \leq \frac{1}{2}$, **never B**.

Fig. 1.3.2: a best reply only to a mixed belief. With B = 2, B is a best reply to neither L (T gives 3) nor R (M gives 3). Against a mixed belief, B beats both iff $2 > 3q$ and $2 > 3(1-q)$, i.e.

$$\tfrac{1}{3} < q < \tfrac{2}{3}.$$

So with pure beliefs only, B would be wrongly eliminated as "never a best response".



Left: $B = 1$ is below the mixture's 3/2 everywhere: dominated by a mix. Right: $B = 2$ rises above the upper envelope of T and M on $(\frac{1}{3}, \frac{2}{3})$ (shaded): a best reply only to mixed beliefs. Source: Gibbons Figs. 1.3.1–1.3.2, pp. 32–33.

The two panels are the two halves of Prop. 6.2: in the left one, "below the upper envelope everywhere" coincides with "below some mixture everywhere"; in the right one, sticking above the envelope somewhere is exactly what survives.

Cabral makes the same point in a footnote (f, p. 162): in his Fig. 7.2 variant, M is not dominated by any pure strategy and is a best reply to the belief that player 2 plays C with probability 2/3 and R with 1/3.

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3 · Worked — Tadelis's 3×3 game (pp. 115–116)

P1 \ P2	L	C	R
U	5, 1	1, 4	1, 0
M	3, 2	0, 0	3, 5
D	4, 3	4, 4	0, 3

No pure domination. Player 2: L beats C against M ($2 > 0$), beats R against U ($1 > 0$), so L is not pure-dominated; C and R are best replies to U and M respectively. Player 1: U is best against L, D against C, M against R. Pure IESDS removes nothing.

Round 1: L is dominated by $\sigma_2 = (0, \frac{1}{2}, \frac{1}{2})$.

Row	$u_2(L)$	$u_2(\frac{1}{2}C + \frac{1}{2}R)$	
U	1	$\frac{1}{2}(4) + \frac{1}{2}(0) = 2$	$2 > 1$
M	2	$\frac{1}{2}(0) + \frac{1}{2}(5) = 2.5$	$2.5 > 2$
D	3	$\frac{1}{2}(4) + \frac{1}{2}(3) = 3.5$	$3.5 > 3$

Remove L.

Round 2: in the reduced game (columns C, R), U is dominated by $\sigma_1 = (0, \frac{1}{2}, \frac{1}{2})$.

Column	$u_1(U)$	$u_1(\frac{1}{2}M + \frac{1}{2}D)$	
C	1	$\frac{1}{2}(0) + \frac{1}{2}(4) = 2$	$2 > 1$
R	1	$\frac{1}{2}(3) + \frac{1}{2}(0) = 1.5$	$1.5 > 1$

Remove U. (U was **not** dominated before L went: against L, U's 5 beats the mix's 3.5. Order matters for *what you can see* at each round, though not for the final survivors.)

What is left is the 2×2 game of §6.2.3 (M, D vs C, R), which has no dominated strategy, pure or mixed, because each of its strategies is a best reply to something. Its three equilibria are in L02-07-mixed-strategies §5.

How to find the dominating mix (p. 116). "A good eye for numbers", or a linear program (L02-12-python §8). Strict dominance is an open condition: shift a small ε of probability and the mix still dominates, so there are infinitely many dominating mixtures (Tadelis exercise 6.3, p. 123, asks for others).

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4 · C — the equivalence

Fact (p. 116). If σ_i is strictly dominated, it is never a best response. (If σ_i' does strictly better against every σ_{-i}, no belief makes σ_i at least as good as σ_i'.)

So every strategy removed by IESDS is removed by rationalizability: **rationalizable ⊆ IESDS survivors**.

Proposition 6.2 (p. 116). In any **two-player** game, a strategy is strictly dominated **iff** it is never a best response. Hence the two procedures give the **same** survivors. Proof beyond the course (Fudenberg and Tirole 1991, Ch. 2; Bernheim 1984, Pearce 1984). With three or more players, rationalizability can be strictly smaller unless beliefs may be correlated (footnote 7, p. 117; Gibbons footnote 13, p. 31).

	IESDS	Rationalizability
Removes	strategies strictly dominated by a pure **or mixed** strategy	strategies that are a best response to **no** (mixed) belief
Two players	same survivors (Prop. 6.2)	same survivors
$n \geq 3$	can be larger	can be smaller
Relation to NE	every NE survives both (L02-02-nash-equilibrium Prop. 5.1)	every NE is rationalizable

---

5 · G — MWG §8.C (pp. 243–245)

MWG builds rationalizability directly on mixed strategies (Defs. 8.C.1–8.C.2, p. 243), remarks that a probabilistic conjecture over rivals' mixed strategies reduces to a deterministic mixed profile when rivals' choices are treated as independent, and proves that the set of rationalizable strategies is no larger than the set surviving iterated strict dominance. Named only.

Links

- Neighbours: L02-01-best-response-rationalizability (pure beliefs), L02-07-mixed-strategies.
- Lecture 1 (pure IESDS, same procedure with fewer dominators): Lecture 1.
- Code: L02-12-python §8 (LP test for dominance by a mixed strategy).
- Practice: Tadelis 6.2, 6.3 (p. 123); MWG 8.C.1 (p. 263); Gibbons 1.2 (p. 48) revisited with mixed dominance.

6.4 · Nash's existence theorem: an equilibrium is a fixed point, and fixed points exist

> **Claim.** A profile is a Nash equilibrium **iff it is a fixed point of the joint best-response correspondence** $BR(\sigma) = BR_1(\sigma_{-1}) \times \cdots \times BR_n(\sigma_{-n})$. In a finite game, mixed strategies make the strategy space a **compact convex simplex** and expected payoffs **linear**, which delivers Kakutani's four conditions; so **every finite game has at least one Nash equilibrium, possibly mixed**. When the conditions fail (asymmetric Bertrand with continuous prices), existence can fail too.

Prev: L02-08-iesds-rationalizability-revisited · Next: L02-10-formulas · Map: L02-00-map

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1 · C — the theorem

Theorem (Nash, 1950; slide 51; Tadelis p. 117). *Every finite normal-form game has a Nash equilibrium in mixed strategies* (slide 51). Tadelis's wording: Any n-player normal-form game with finite strategy sets S_i for all players has a Nash equilibrium in mixed strategies. (A pure equilibrium counts: pure strategies are degenerate mixtures, footnote 8.)

Gibbons's statement (p. 45): in $G = \{S_1, \ldots, S_n; u_1, \ldots, u_n\}$, if n is finite and each S_i is finite, there exists at least one NE, possibly in mixed strategies.

> **Sufficient, not necessary.** Cournot and the commons have infinite strategy sets and still have equilibria (constructed by hand, "following Cournot's lead", Gibbons p. 11). The theorem guarantees existence where we cannot construct.

> **Not covered in class from here on.** Slide 51 states the theorem without proof; §2–§6 are reading.

2 · U — Brouwer in one dimension (p. 117, Fig. 6.5)

Brouwer's Fixed-Point Theorem. If $f : [0,1] \to [0,1]$ is continuous, some x^* has $f(x^*) = x^*$.

Why. If $f(0) = 0$ or $f(1) = 1$ we are done. Otherwise $f(0) > 0$ and $f(1) < 1$: f starts **above** the 45° line and ends **below** it, so the continuous function $g(x) = f(x) - x$ goes from positive to negative and has a zero (the intermediate value theorem). A function that **jumps** across the diagonal (f_3 in Fig. 6.5) need not cross it.

Each hypothesis matters (footnote 10, p. 121): on $(0,1)$ (not closed) $f(x) = \sqrt{x}$ stays above the diagonal with no fixed point; on $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$ (not convex) $f = \frac{3}{4}$ on the first piece and $\frac{1}{4}$ on the second jumps over the gap.

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3 · C — equilibrium as a fixed point (pp. 118–120)

Definition 6.9 (p. 119). The collection of best-response correspondences $BR \equiv BR_1 \times \cdots \times BR_n$ maps the set of mixed profiles $\Delta S = \Delta S_1 \times \cdots \times \Delta S_n$ into its subsets: $BR : \Delta S \rightrightarrows \Delta S$.

Fact (p. 120). σ^* is a Nash equilibrium **iff** $\sigma^* \in BR(\sigma^*)$.

Why. $\sigma^* \in BR(\sigma^*)$ says $\sigma_i^* \in BR_i(\sigma_{-i}^*)$ for every i, which is Def. 6.6 word for word.

Worked — the §6.2.3 game (Fig. 6.6). With $BR(q,p) = (BR_2(p), BR_1(q))$ and the correspondences of L02-07-mixed-strategies §5:

Point (q,p)	$BR_1(q)$	$BR_2(p)$	$BR(q,p)$	Fixed point?
$(0,1)$	$q < \frac{3}{7} \Rightarrow p = 1$	$p > \frac{1}{6} \Rightarrow q = 0$	$(0,1)$	**yes**: (M, R)
$(1,0)$	$q > \frac{3}{7} \Rightarrow p = 0$	$p < \frac{1}{6} \Rightarrow q = 1$	$(1,0)$	**yes**: (D, C)
$(\frac{3}{7}, \frac{1}{6})$	$[0,1]$	$[0,1]$	$[0,1]^2$	**yes**: it contains itself
$(0.8, 0.5)$	$p = 0$	$q = 0$	$(0,0)$	no
$(0.2, 0.1)$	$p = 1$	$q = 1$	$(1,1)$	no

---

4 · C — Kakutani, and why mixed strategies satisfy it (pp. 120–122)

Theorem 6.1 (Kakutani's Fixed-Point Theorem, p. 120). A correspondence $C : X \rightrightarrows X$ has a fixed point $x \in C(x)$ if (1) X is a non-empty, compact and convex subset of $\mathbb{R}^n$; (2) $C(x)$ is non-empty for all x; (3) $C(x)$ is convex for all x; (4) C has a closed graph.

Definitions (p. 120): **convex**: $\lambda x + (1-\lambda)y \in X$ for $x, y \in X$, $\lambda \in [0,1]$; **closed**: limits of sequences in X stay in X ($(0,1]$ is not closed: $1/n \to 0$); **compact**: closed and bounded; **closed graph**: if $x_n \to x$, $y_n \in C(x_n)$ and $y_n \to y$, then $y \in C(x)$.

The four conditions for any 2×2 game (pp. 121–122).

Condition	Why it holds
(1) $X = [0,1]^2$	non-empty, convex, compact
(2) BR non-empty	expected payoff is continuous (linear) in own mix on the compact $[0,1]$: a maximum exists (extreme value theorem)
(3) BR convex-valued	by Prop. 6.1, if two mixes are best replies the player is indifferent between the pure strategies they use, so **every** mix between them is a best reply; BR_i is $\{0\}$, $\{1\}$ or $[0,1]$
(4) closed graph	if $q_n \in BR_2(p_n)$, i.e. $u_2(q_n, p_n) \geq u_2(q, p_n)$ for all q, and $(q_n, p_n) \to (\hat{q}, \hat{p})$, continuity lets the inequality pass to the limit: $u_2(\hat{q}, \hat{p}) \geq u_2(q, \hat{p})$, so $\hat{q} \in BR_2(\hat{p})$

So BR has a fixed point and **every 2×2 game has a NE**. With m pure strategies the simplex $\Delta_i = \{\sigma_i : \sigma_i(s_{ik}) \in [0,1], \sum_k \sigma_i(s_{ik}) = 1\}$ is non-empty, compact and convex in $\mathbb{R}^m$, and the same four arguments go through (pp. 122–123).

> **Mixed strategies are what make existence work.** A finite set of pure strategies is not convex, so condition (3) fails for pure best replies (Matching Pennies has no pure fixed point). Allowing mixtures convexifies the strategy space and turns indifference into a convex set of best replies (MWG footnote 5, p. 253).

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5 · C — when existence fails: back to Bertrand

Asymmetric Bertrand with continuous prices (L02-04-bertrand §4) has infinite strategy sets **and** a payoff that jumps at $p_1 = p_2$, so firm 1's best-response set to $p_2 = 2$ is **empty**: condition (2) fails. Tadelis's choice (p. 93) is to avoid such discontinuities, or restore discreteness, rather than to abandon NE.

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6 · G — MWG's general statements

- **Prop. 8.D.2 (p. 252):** every game whose strategy sets $S_1, \dots, S_I$ are finite has a mixed-strategy NE.
- **Prop. 8.D.3 (p. 253):** a (pure) NE exists if each S_i is a non-empty, convex, compact subset of some $\mathbb{R}^M$ and u_i is **continuous** in all strategies and **quasiconcave** in s_i. Cournot and the commons satisfy it; homogeneous Bertrand violates continuity.
- **Appendix A (pp. 260–261):** Lemma 8.AA.1 shows $b(\cdot)$ is non-empty, convex-valued and upper hemicontinuous; Kakutani then gives Prop. 8.D.3, and Prop. 8.D.2 follows by applying 8.D.3 to the mixed extension.

Not examinable unless the lecture states it.

Links

- Neighbours: L02-07-mixed-strategies (Prop. 6.1, used for convexity), L02-04-bertrand (non-existence).
- Lecture 7 (existence in Bayesian games reuses the same fixed-point idea): Lecture 7.
- Code: L02-12-python §9 (fixed points of BR on a grid).
- Practice: Gibbons 1.10 (pp. 50–51); Tadelis 6.10 (p. 125, the all-pay auction with no finite-support equilibrium).

Lecture 2 · Formulas: every definition, result and number

Prev: L02-09-existence · Next: L02-11-pitfalls · Map: L02-00-map

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0 · The exam specification, mapped

The specification is the slides (`Aula/Lecture02.pdf`, Tue Oct 20 2026), item by item. Rows 7–10 are the book's material that the lecture did not teach: reading, examinable only if a problem set brings it.

#	What to be able to do	Slides	Note
1	Define a best response, the best-response correspondence, "never a best response"; describe rationalizability; state Lessons 3–4	9–12	L02-01-best-response-rationalizability
2	State the NE definition and its three interpretations; prove Props. 1–2 (NE survives IESDS; unique IESDS survivor is the NE); find pure NE in a matrix (road game, PD, BoS)	14–18	L02-02-nash-equilibrium
3	Cournot: payoffs, benchmarks q^m and q^{cp}, $q_i^*(q_j) = q^m - \frac{1}{2}q_j$, NE $\frac{2}{3}q^m$, the IESDS sequence, the cartel as a PD, n firms	20–27	L02-03-cournot
4	Bertrand: $p^m = \frac{a+c}{2}$, the BR correspondence, (c, c) is the unique NE (two steps), the paradox and its assumptions, Cournot or Bertrand	29–40	L02-04-bertrand
5	Mixed strategies, expected payoffs, mixed NE; the principle of indifference with its proof; all NE of the BoS; interpretations; Nash's theorem	42–51	L02-07-mixed-strategies

#	What to be able to do	Slides	Note
6	Mini case (monitoring): no pure NE, mixed NE, comparative statics (graded; method only here)	53–55	L02-07-mixed-strategies §9
7	*Reading:* tragedy of the commons	—	L02-05-commons
8	*Reading:* Hotelling voting and the median voter	—	L02-06-hotelling-voting
9	*Reading:* IESDS with mixed dominance; Prop. 6.2	—	L02-08-iesds-rationalizability-revisited
10	*Reading:* the fixed-point proof of existence (Kakutani for 2×2 games)	—	L02-09-existence

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1 · Definitions and results (book numbering)

Label	Statement	Note
Def. 4.5 (p. 70)	s_i is a best response to s_{-i} if $u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i})\ \forall s_i'$	01 §2
Claim 4.2 (p. 70)	a rational player best-responds to his belief	01 §2
Prop. 4.3 (p. 70)	strictly dominated ⇒ never a best response (to a pure belief)	01 §2
Prop. 4.4 (p. 70)	strict DSE or unique IESDS survivor ⇒ each s_i^* is a BR to s_{-i}^*	01 §2
Def. 4.6 (p. 72)	belief = profile $s_{-i} \in S_{-i}$	01 §2
Def. 4.7 (p. 72)	$BR_i(s_{-i}) \subseteq S_i$, a correspondence	01 §2
Def. 4.8 (p. 73)	never a best response: no belief makes s_i a BR	01 §2
Def. 5.1 (p. 80)	NE: $u_i(s_i^*, s_{-i}^*) \geq u_i(s_i', s_{-i}^*)\ \forall s_i', \forall i$	02 §2
Prop. 5.1 (pp. 80–81)	strict DSE, unique IESDS survivor or unique rationalizable profile ⇒ unique NE	02 §2
Gibbons App. 1.1.C (pp. 12–13)	Prop. A: unique IESDS survivor is the unique NE; Prop. B: every NE survives IESDS	02 §2

Label	Statement	Note
Slides' Prop. 1 (slide 16)	NE ⇒ survives IESDS (proof: first round of elimination)	02 §2c
Slides' Prop. 2 (slide 17)	finite game, unique IESDS survivor ⇒ NE (proof: chain of dominators)	02 §2c
Slide 16 note	a NE may use weakly dominated strategies (Bertrand $p_i^* = c$)	02 §2c, 04 §0
eq. (5.1) (p. 85)	$u_i = \ln k_i + \ln(K - \sum_j k_j)$	05 §2
eq. (5.2) (p. 87)	$BR_i(q_j) = (a - bq_j - c_i)/(2b)$	03 §1
Prop. 5.2 (p. 91)	continuous homogeneous Bertrand, equal costs: unique NE $p_1 = p_2 = c$	04 §3
Def. 6.1 (p. 102)	mixed strategy $\sigma_i \in \Delta S_i$	07 §2
Def. 6.2 (p. 104)	support: $\sigma_i(s_i) > 0$	07 §2
Def. 6.3 (p. 104)	mixed strategy on an interval: a CDF F_i	07 §2
Def. 6.4 (p. 105)	belief $\pi_i \in \Delta S_{-i}$	07 §2
Def. 6.5 (p. 105)	$u_i(s_i, \sigma_{-i}) = \sum_{s_{-i}} \sigma_{-i}(s_{-i}) u_i(s_i, s_{-i})$; $u_i(\sigma_i, \sigma_{-i}) = \sum_{s_i} \sigma_i(s_i) u_i(s_i, \sigma_{-i})$	07 §2
Def. 6.6 (p. 107)	mixed NE: $u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*)\ \forall \sigma_i$	07 §2
Prop. 6.1 (p. 108)	pure strategies in the support of an equilibrium mix earn equal payoffs	07 §2
Principle of indifference (slides 44–46)	σ^* NE ⇔ (i) $u_i(\sigma^*) = u_i(s_i, \sigma_{-i}^*)$ on $\operatorname{supp} \sigma_i^*$; (ii) $u_i(\sigma^*) \geq u_i(s_i, \sigma_{-i}^*)$ off it	07 §2b
Claims 6.1–6.2 (p. 112)	RPS: no NE with a pure player; none with a two-strategy mix	07 §4
Def. 6.7 (p. 114)	s_i strictly dominated by σ_i: $u_i(\sigma_i, s_{-i}) > u_i(s_i, s_{-i})\ \forall s_{-i}$	08 §1
Def. 6.8 (p. 114)	σ_i never a BR: no σ_{-i} with $\sigma_i \in BR_i(\sigma_{-i})$	08 §1
Fact (p. 116)	strictly dominated ⇒ never a BR	08 §4
Prop. 6.2 (p. 116)	two players: strictly dominated ⇔ never a BR	08 §4
Nash's theorem (p. 117)	finite games have a mixed-strategy NE	09 §1
Brouwer (p. 117)	continuous $f : [0,1] \to [0,1]$ has a fixed point	09 §2
Def. 6.9 (p. 119)	$BR = BR_1 \times \cdots \times BR_n : \Delta S \rightrightarrows \Delta S$	09 §3
Fact (p. 120)	σ^* is a NE ⇔ $\sigma^* \in BR(\sigma^*)$	09 §3

Label	Statement	Note
Thm. 6.1 Kakutani (p. 120)	non-empty compact convex X; $C(x)$ non-empty, convex; closed graph ⇒ fixed point	09 §4
Gibbons (p. 42)	2×2 cases (i)–(iv); $q' = (w-y)/((x-z)+(w-y))$	07 §6
MWG Def. 8.C.1–8.C.2 (p. 243)	never a BR; rationalizable strategies (G)	01 §6, 08 §5
MWG Def. 8.D.1 (p. 246), 8.D.2 and Prop. 8.D.1 (p. 250)	pure NE; mixed NE; support characterisation (G)	02 §6, 07 §8
MWG Props. 8.D.2 (p. 252), 8.D.3 (p. 253); App. A (pp. 260–261)	existence, finite and continuous-quasiconcave (G)	09 §6

1.1 · Closed forms

Model	Result
Cournot, slides' notation	$q^m = \frac{a-c}{2b}$, $q^{cp} = \frac{a-c}{b} = 2q^m$; $q_i^*(q_j) = q^m - \frac{1}{2}q_j$ (0 if $q_j \geq q^{cp}$); NE $q_i^* = \frac{2}{3}q^m$; IESDS bounds $\frac{1}{2}, \frac{3}{4}, \frac{5}{8}, \frac{11}{16}, \ldots \to \frac{2}{3}$ (times q^m)
Cournot, n firms (slides)	$q_i^* = 2q^m - Q^*$; $Q^* = \frac{2n}{n+1}q^m \to q^{cp}$; $q_i^* = \frac{2q^m}{n+1}$
Bertrand, slides	$Q(p) = \frac{a-p}{b}$; $p^m = \frac{a+c}{2}$; $p_i^*(p_j)$: $\{p_i > p_j\}$, $\{p_i \geq p_j\}$, $\emptyset$, $\{p^m\}$ for $p_j < c$, $= c$, $\in (c, p^m]$, $> p^m$; unique NE (c, c)
BoS mixed (slides 47–50)	$u_A(O,q) = 2q$, $u_A(F,q) = 1-q \Rightarrow q = \frac{1}{3}$; $u_C(O,p) = p$, $u_C(F,p) = 2-2p \Rightarrow p = \frac{2}{3}$; payoffs $\frac{2}{3}$ each
Cournot, $P = a - bQ$, c_i	$q_i^* = \frac{a-2c_i+c_j}{3b}$; $Q^* = \frac{2a-c_1-c_2}{3b}$; $P^* = \frac{a+c_1+c_2}{3}$; $\pi_i^* = b(q_i^*)^2$; firm i out iff $c_i \geq (a+c_j)/2$
Cournot, n symmetric firms	$q_i^* = \frac{a-c}{(n+1)b}$, $P^* = \frac{a+nc}{n+1} \to c$
Collusion temptation	$BR(q_m/2) = \frac{3(a-c)}{8b} > \frac{a-c}{4b} = \frac{q_m}{2}$
Bertrand, homogeneous	$p_1 = p_2 = c$ (continuous); $\{(c,c), (c+\Delta, c+\Delta)\}$ on a grid
Bertrand, differentiated $q_i = a - p_i + bp_j$	$BR_i = (a+bp_j+c)/2$; $p^* = (a+c)/(2-b)$, $b < 2$
Commons, 2 players	$BR_i = (K-k_j)/2$; NE $K/3$; optimum $K/4$; gain $\ln(9/8)$ each
Commons, Gibbons	NE: $v(G^*) + \frac{1}{n}G^*v'(G^*) = c$; optimum: $v(G^{**}) + G^{**}v'(G^{**}) = c$; $G^* > G^{**}$
Hotelling voting	$BR_i(a_j) = [-a_j + 1, a_j - 1]$ for $a_j > 0$; NE $a_1 = a_2 =$ median
2×2 mixed NE	q^* solves $u_1(\text{row}_1, q) = u_1(\text{row}_2, q)$; p^* solves $u_2(p, \text{col}_1) = u_2(p, \text{col}_2)$

2 · Every number in the lecture's reading

Example (page)	Numbers
Slides Cournot example (20–25)	$q^m = 45$, $q^{cp} = 90$; $q_i^*(q_j) = \frac{90-q_j}{2}$; NE 30, $Q^* = 60$, $P^* = 40$, $\Pi^* = 900$; cartel 22.5 each, 1,012; deviation 33.75 earns 1,139, cartel firm 759; (Deviate, Deviate) printed 900
Slides road game, PD (18)	(L, L), (R, R), IESDS removes nothing; PD $(2,2), (-1,3), (3,-1), (0,0)$: unique NE (D, D)
Slides BoS (9, 14, 47–50)	(O, O), (F, F), $(p, q) = (\frac{2}{3}, \frac{1}{3})$; payoffs $\frac{2}{3}$; miscoordination $\frac{5}{9}$
Slides mini case (54)	$w = 10$, $g = 2$, $v = 20$, $h = 1$; then $h = 0.5$, $w = 20$ (graded: not solved here)
BR with ties (p. 72)	$BR_1(R) = \{U, D\}$, $BR_1(L) = \{M, D\}$
Cournot rationalizability (pp. 73–74)	$BR = (90 - q)/2$; rounds $[0, 45]$, $[22.5, 45]$, $[22.5, 33.75]$, … $\to 30$
Beauty contest (pp. 74–75)	1, 5, 18 $\to$ average 8, target 6, player 2 wins; 20 never a BR; 19 rounds $\to \{0, 1\}$ ($n > 2$), $\{0\}$ ($n = 2$)
Underline method (p. 81)	P1 BR at (D,L), (M,C), (M,R); P2 at (U,R), (M,C), (D,C); unique pure NE (M, C); four more NE in mixed strategies, e.g. $(\frac{1}{6}, \frac{5}{6}, 0)$ for both
Stag Hunt (p. 84)	(S,S) 5,5 and (H,H) 3,3 both NE
Commons (pp. 86–87)	$K/3$ vs $K/4$; $\Delta v = 2\ln 3 - \ln 2 - \ln 4 = 0.118$ (book prints 0.051, the base-10 value)
Cournot (pp. 87–90)	$q_i = 30$, $P = 40$, $\pi_i = 900$; monopoly 45, 55, 2025; competitive 90 at 10; deviation $BR(22.5) = 33.75$
Bertrand (pp. 89–91)	monopoly $p = 55$; $\Delta v = 55.02p_j - 0.5p_j^2 - 501.1$, 0.0002 at 10.02 (exact 0.0001); NE (10,10), (10.01,10.01); profits 0 or about 0.45
Asymmetric Bertrand (pp. 91–92)	$c_1 = 1$, $c_2 = 2$: (1.99, 2.00) is a NE on cents; continuous: $u_1(p_1, 2) \to 98$, $= 49$ at 2; $p_1^M = 50.5$
Hotelling (p. 94)	$a_1 = -15$, $a_2 = 22$: citizen 3 votes 1 (18 vs 19), citizen 4 votes 2; 54 vs 47
Matching Pennies (pp. 108–111)	$u_1(H, q) = 2q - 1$, $u_1(T, q) = 1 - 2q$; NE $(\frac{1}{2}, \frac{1}{2})$
RPS (pp. 106, 111–113)	vs $(\frac{1}{2}, \frac{1}{2}, 0)$: R $-\frac{1}{2}$, P $\frac{1}{2}$, S 0; NE $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
All-pay auction (p. 107)	vs $U[0, 1]$: every bid in $[0, 1]$ earns 0
§6.2.3 game (pp. 113–114)	pure (M,R), (D,C); mixed $p = \frac{1}{6}$, $q = \frac{3}{7}$; payoffs $\frac{12}{7}$, $\frac{10}{3}$
Mixed IESDS (pp. 115–116)	L vs $(0, \frac{1}{2}, \frac{1}{2})$: 1<2, 2<2.5, 3<3.5; then U vs $(0, \frac{1}{2}, \frac{1}{2})$: 1<2, 1<1.5
Gibbons Figs. 1.3.1–1.3.2 (pp. 32–33)	$\frac{1}{2}T + \frac{1}{2}M = \frac{3}{2} > 1$; B = 2 is a BR iff $\frac{1}{3} < q < \frac{2}{3}$
Battle of the Sexes mixed (Gibbons p. 40)	each attends own favourite with prob. $\frac{2}{3}$; miscoordination $\frac{5}{9}$

Lecture 2 · Pitfalls and a True/False self-test

Prev: L02-10-formulas · Next: L02-12-python · Map: L02-00-map

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Part I — the confusions this material produces

1 · "One player best-responds, so it is an equilibrium"

Bertrand with $p_2 = 10$: every $p_1 \geq 10$ is a best reply for firm 1. But $(12, 10)$ is **not** a NE: firm 2's price of 10 is not a best reply to 12 (it would rather price at 11.99). Equilibrium is **mutual** best response (Tadelis p. 90). → L02-04-bertrand §2.

2 · "Rationalizable (or IESDS-surviving) means Nash"

Battle of the Sexes: (O, F) survives both and is not a NE; each player best-responds to a **wrong** belief. The inclusion runs one way only: NE ⊆ rationalizable ⊆ IESDS survivors. → L02-02-nash-equilibrium §2.

3 · "A Nash equilibrium is efficient" / "the Pareto-best equilibrium will be played"

Prisoner's dilemma, Cournot (1800 total vs 2025), the commons ($K/3$ vs $K/4$): NE need not be Pareto optimal. Stag Hunt: (H, H) is a NE although (S, S) Pareto dominates it; nothing in the definition selects (S, S). → L02-05-commons, L02-02-nash-equilibrium §4.

4 · "In a mixed equilibrium my mix makes *me* indifferent"

It makes the **other** player indifferent. In the §6.2.3 game player 1's mix $p = \frac{1}{6}$ comes from **player 2's** payoffs ($4(1-p) = 3 + 2p$); change player 1's payoffs (keeping the best-response pattern) and p does not move, q does. → L02-07-mixed-strategies §3.

5 · "Indifferent between two strategies, so any mix of all strategies is fine"

Prop. 6.1 covers strategies **in the support**; MWG Prop. 8.D.1(ii) adds that strategies **outside** it must not pay more. A mix that puts weight only on equal-payoff strategies is a best reply only if no other strategy pays strictly more. → L02-07-mixed-strategies §8.

6 · "Not dominated by any pure strategy, so it survives IESDS"

Tadelis p. 115: L is beaten by $(0, \frac{1}{2}, \frac{1}{2})$ though by neither C nor R alone; Gibbons's B (1, 1) is beaten by $\frac{1}{2}T + \frac{1}{2}M$. → L02-08-iesds-rationalizability-revisited.

7 · "Never a best response to a pure belief means never a best response"

Gibbons Fig. 1.3.2: B = (2, 2) is a best reply to neither L nor R but is the unique best reply for $q \in (\frac{1}{3}, \frac{2}{3})$. Beliefs must be allowed to be mixed. → L02-08-iesds-rationalizability-revisited §2.

8 · "A best-response function always exists"

Asymmetric Bertrand with continuous prices: against $p_2 = 2$ firm 1's profit approaches 98 but never reaches it, so $BR_1(2) = \emptyset$. Correspondences can be multi-valued (ties, p. 72) **or empty** (discontinuities). → L02-04-bertrand §4.

9 · "Nash's theorem guarantees a pure equilibrium" / "no equilibrium without finiteness"

Nash's theorem: finite games, **mixed** equilibrium (Matching Pennies has no pure one). It is sufficient, not necessary: Cournot and the commons have infinite strategy sets and pure equilibria. → L02-09-existence.

10 · "Cournot equilibrium" and "Bertrand equilibrium" are different solution concepts

They are the **same** concept, Nash, applied to two different games (quantities vs prices) (Gibbons p. 21).

11 · Substitutes vs complements, read off the wrong axis

Strategic substitutes/complements describe the slope of the **best response** (Cournot down, Bertrand up, p. 93), not whether the **products** are substitutes (they are substitutes in both models).

12a · "A Nash equilibrium never uses a dominated strategy"

Only **strictly** dominated strategies are excluded (slides' Prop. 1). Bertrand's $p_i^* = c$ is **weakly** dominated, yet (c, c) is the unique NE (slides 16, 36). → L02-02-nash-equilibrium §2c.

12b · "In the mixed BoS equilibrium each player is better off than at a pure one, since nobody is left out"

Each expects $\frac{2}{3}$, **less than in either pure equilibrium** (the worse-off player gets 1 there): they miscoordinate with probability $\frac{5}{9}$ (slide 50). → L02-07-mixed-strategies §2c.

12c · Reading the cartel matrix's (Deviate, Deviate) cell as "both produce 33.75"

The printed 900 is the payoff at the Nash outputs (30, 30). Both at 33.75 would give each 759.38, a tie with staying in the cartel (slide 25). → L02-03-cournot §4b.

12 · Two printed slips worth knowing

Tadelis p. 87 footnote 5 gives the commons gain as 0.051, which is the **base-10** value; with the natural log of eq. (5.1) it is $\ln(9/8) \approx 0.118$. Tadelis p. 93 (Remark) calls the monopoly price "45": it is **55** (45 is the quantity). Neither changes a conclusion.

Part II — True/False self-test

Original items in the `macro-1` style: one claim per item. Score $= \max\{0, \textit{right} - \textit{wrong}/2\}$.

1. **If a game has a unique profile surviving IESDS, that profile is the game's unique Nash equilibrium. True.** Prop. 5.1: the survivor is a mutual best reply (Prop. 4.4), and every NE survives IESDS, so no other NE can exist.
2. **In the 3×3 game of Tadelis p. 81, (M, C) is the only Nash equilibrium. False.** It is the only **pure** one (which is all Tadelis claims). Support enumeration finds four more, all mixed: e.g. both players put $\frac{1}{6}$ on their first strategy and $\frac{5}{6}$ on their second; then U and M each pay player 1 $\frac{27}{6} = 4.5$ and D pays $\frac{23}{6}$, and symmetrically for player 2. Five equilibria in all, an odd number as footnote 6 (p. 113) leads one to expect (L02-12-python §7).
3. **A strictly dominant strategy is a best response to every belief, pure or mixed. True.** It beats each alternative against every pure profile, hence against every average of them.

4. **A pure strategy that is not a best response to any pure strategy of the opponent can be a best response to some mixed strategy. True.** Gibbons Fig. 1.3.2: B = (2, 2) is optimal for $q \in (\frac{1}{3}, \frac{2}{3})$.

5. **In a two-player game, a strategy is never a best response if and only if it is strictly dominated, possibly by a mixed strategy. True.** Prop. 6.2; the "possibly mixed" is essential.

6. **In a mixed-strategy Nash equilibrium, each player's equilibrium mixing probabilities are determined by that player's own payoffs. False.** They are pinned by the **opponent's** indifference condition (Pitfall 4).

7. **If a player mixes between two pure strategies in a Nash equilibrium, both earn him the same expected payoff, equal to his equilibrium payoff. True.** Prop. 6.1.

8. **Matching Pennies has a Nash equilibrium. True.** $(\frac{1}{2}, \frac{1}{2})$ for both; it has no **pure** one.

9. **In the Cournot duopoly with $P = a - bQ$ and equal costs c, each firm's equilibrium output is half the monopoly output. False.** $q_i^* = (a-c)/(3b)$, while half the monopoly output is $(a-c)/(4b)$: the duopoly overproduces relative to collusion because $BR(q_m/2) > q_m/2$.

10. **In the linear Cournot duopoly, a one-unit increase in firm 1's marginal cost raises the equilibrium price by one-third. True.** $P^* = (a + c_1 + c_2)/3$, so $\partial P^*/\partial c_1 = \frac{1}{3}$ (interior case).

11. **In the homogeneous-good Bertrand duopoly with equal constant marginal costs and continuous prices, both firms earn zero profit in equilibrium. True.** Prop. 5.2: $p_1 = p_2 = c$.

12. **In the homogeneous Bertrand duopoly with continuous prices, if firm 1's marginal cost is 1 and firm 2's is 2, then $p_1 = p_2 = 2$ is a Nash equilibrium. False.** Firm 1 would rather price just below 2; and there is no best such price, so no pure NE exists (pp. 91–93).

13. **Cournot competition is a game of strategic substitutes and Bertrand competition with differentiated products is a game of strategic complements. True.** Best responses slope $-1/2$ and $+b/2$ respectively.

14. **In the two-player tragedy of the commons of Tadelis §5.2.2, both players would be strictly better off if each consumed less than in the Nash equilibrium. True.** At $K/4$ each gains $\ln(9/8) > 0$ relative to $K/3$.

15. **In the Hotelling voting game with 101 citizens, the best response to a rival at platform $a_j = +10$ is unique. False.** It is every platform in $[-9, 9]$; only against $a_j \in \{-1, 0, 1\}$ is it unique (0).

16. **A Nash equilibrium always exists in pure strategies when each player has finitely many strategies. False.** Matching Pennies and rock-paper-scissors; Nash's theorem guarantees a **mixed** one.

17. **If a game has two pure-strategy Nash equilibria, it must also have a mixed-strategy Nash equilibrium. False.** "Almost always", not always (Tadelis footnote 6, p. 113; Gibbons footnote 15, p. 45). Counterexample: both players get 1 if both play their first strategy and 0 otherwise. (first, first) and (second, second) are NE, but a player is indifferent only if the other plays the second strategy for sure, so no profile with a mixing player is an equilibrium. The ties (weak dominance) are what break the generic rule.

18. **A Nash equilibrium is a fixed point of the players' joint best-response correspondence. True.** Tadelis's Fact, p. 120.

19. **Kakutani's theorem needs the best-response correspondence to be convex-valued; in finite games this is guaranteed by allowing mixed strategies. True.** Indifference between pure strategies makes every mix between them a best reply (Prop. 6.1); pure best-reply sets are not convex.

20. **If Chris believes Alex will go to the opera, Chris's unique best response in the Battle of the Sexes is to go to the opera, so (O, O) is the unique Nash equilibrium. False.** The first half is true ($u_2(O, O) = 1 > 0$); the conclusion is not: (F, F) and the mixed profile are also NE.

21. **In the all-pay auction for a dollar, if the rival bids uniformly on $[0, 1]$, every bid in $[0, 1]$ gives the same expected payoff. True.** Each gives 0 (p. 107).

22. **The p-beauty contest prediction that nobody chooses a number above 1 relies only on each player being rational. False.** It needs **common knowledge** of rationality, 19 rounds deep; rationality alone removes only 20.

23. **Rationalizability always selects a unique outcome in the Battle of the Sexes. False.** "Anything can happen" (p. 76).

24. **In the mixed equilibrium of Tadelis's §6.2.3 game, player 1 plays M with probability $\frac{3}{7}$. False.** $\frac{3}{7}$ is player 2's probability of C; player 1 plays M with probability $\frac{1}{6}$ (Pitfall 4 again).

Items 25–30 follow the slides.

25. **Every Nash equilibrium survives the iterated elimination of strictly dominated strategies. True.** Slides' Prop. 1: at the first round some s_i^* fell, its dominator would beat it against s_{-i}^*.

26. **A Nash equilibrium cannot involve a weakly dominated strategy. False.** Bertrand's (c, c) (slides 16, 36).

27. **In the linear Cournot duopoly, each firm's equilibrium output is two-thirds of the monopoly output. True.** $q_i^* = \frac{2}{3}q^m$ (slide 24); total output is $\frac{4}{3}q^m$.

28. **In the Cournot oligopoly with n identical firms, total output equals $\frac{2n}{n+1}$ times the monopoly output. True.** Slide 27; it tends to $2q^m = q^{cp}$.

29. **In the homogeneous Bertrand duopoly with marginal cost $c < p_j \leq p^m$, firm i's best response is to price one cent below p_j. False.** With continuous prices the best-response set is **empty** (slide 34): any price below p_j can be raised slightly and still undercut.

30. **By the principle of indifference, in a mixed Nash equilibrium every pure strategy of a player, including those played with probability zero, yields the same expected payoff. False.** Only strategies **in the support** tie; those outside it must yield **no more** (slide 44, condition 2).

Lecture 2 · Computation in Python

> Every routine the lecture needs, each with `assert` self-checks against the book's own numbers. Run the blocks in order in one session (they share imports); all of them were run together on 2026-10-16 and pass. numpy and scipy only (scipy for the linear program in §8).
>
> Prev: L02-11-pitfalls · Map: L02-00-map · Formulas: L02-10-formulas

Reuse. Lecture 1's checks (`Listas/lista-lecture-01-check.py`) already contain pure-strategy IESDS (`iesds2`) and the Cournot best response and IESDS recursion (`BR`, `cournot_iesds`). They are cited rather than imported because that file runs the whole Lecture 1 list check (sympy, 1500 random draws) on import; the two-line best response is repeated in §3. Dominance **by a mixed strategy** (§8) is new: `iesds2` compares pure strategies only. The figure script (`figures/make_figures.py`) and the companion checker (`companions/check_companions.py`) carry their own copies of the Bertrand, voting and 2×2 routines, so each artefact runs on its own.

---

1 · Best responses and pure equilibria in a matrix → L02-02-nash-equilibrium

```
import itertools
from fractions import Fraction as Fr
import numpy as np


def best_responses(U1, U2):
    """Pure best-response sets of a bimatrix game.

    U1[r][c], U2[r][c]: payoffs of the row and column player.
    Returns (br1, br2): br1[c] = set of rows that are best replies to column c;
    br2[r] = set of columns that are best replies to row r.
    """
    U1, U2 = np.asarray(U1, float), np.asarray(U2, float)
    br1 = [set(np.flatnonzero(U1[:, c] == U1[:, c].max())) for c in
 ↪  range(U1.shape[1])]
    br2 = [set(np.flatnonzero(U2[r, :] == U2[r, :].max())) for r in
 ↪  range(U2.shape[0])]
    return br1, br2


def pure_nash(U1, U2):
    """Tadelis's underline method (pp. 81-82): cells where both players
     ↪  best-respond."""
    br1, br2 = best_responses(U1, U2)
    return [(r, c) for r in range(len(U1)) for c in range(len(U1[0])) if r in br1[c]
     ↪  and c in br2[r]]


# p. 81: no dominated strategy, unique pure NE (M, C)
U1 = [[7, 4, 1], [2, 5, 2], [8, 3, 0]]
U2 = [[7, 2, 8], [4, 5, 3], [1, 2, 0]]
assert pure_nash(U1, U2) == [(1, 1)]
```

```
# p. 72: ties -> BR is a correspondence: BR1(R) = {U, D}, BR1(L) = {M, D}
br1, _ = best_responses([[3, 5, 6], [4, 8, 3], [4, 9, 6]], [[3, 1, 2], [1, 4, 6], [0,
 ↪  6, 8]])
assert br1[2] == {0, 2} and br1[0] == {1, 2}
# p. 80 (Lecture 1's IESDS game): unique NE (U, L)
assert pure_nash([[4, 5, 6], [2, 8, 3], [3, 9, 2]], [[3, 1, 2], [1, 4, 6], [0, 6,
 ↪  8]]) == [(0, 0)]
# Battle of the Sexes p. 69: two pure NE; prisoner's dilemma p. 48: (F, F) only
assert pure_nash([[2, 0], [0, 1]], [[1, 0], [0, 2]]) == [(0, 0), (1, 1)]
assert pure_nash([[-2, -5], [-1, -4]], [[-2, -1], [-5, -4]]) == [(1, 1)]
# Stag Hunt p. 84: (S, S) and (H, H); Matching Pennies p. 101: none
assert pure_nash([[5, 0], [3, 3]], [[5, 3], [0, 3]]) == [(0, 0), (1, 1)]
assert pure_nash([[1, -1], [-1, 1]], [[-1, 1], [1, -1]]) == []
```

---

2 · Rationalizability in Cournot and the beauty contest → L02-01-best-response-rationalizability

```
def cournot_rationalizable(a=100.0, b=1.0, c=10.0, rounds=40):
    """Interval of quantities surviving k rounds (Sec. 4.3.4): [BR(upper),
     ↪  BR(lower)]."""
    br = lambda q: max(0.0, (a - c - b * q) / (2 * b))
    lo, hi = 0.0, br(0.0)                       # round 1: [0, 45]
    hist = [(lo, hi)]
    for _ in range(rounds - 1):
        lo, hi = br(hi), br(lo)
        hist.append((lo, hi))
    return hist


h = cournot_rationalizable()
assert h[0] == (0.0, 45.0) and h[1] == (22.5, 45.0) and h[2] == (22.5, 33.75)
assert abs(h[-1][0] - 30) < 1e-9 and abs(h[-1][1] - 30) < 1e-9   # converges to 30
 ↪  (p. 74)


def beauty_never_best(n, top):
    """Is `top` never a best response in the 3/4-beauty contest on {0..top}? (p. 75).

    Checks the binding belief (everyone else plays `top`): `top - 1` wins alone,
 ↪  `top` only ties.
    """
    others = [top] * (n - 1)
    target = lambda x: 0.75 * (sum(others) + x) / n
    return abs(top - 1 - target(top - 1)) < abs(top - target(top - 1))


assert all(beauty_never_best(n, 20) for n in range(2, 50))
avg = lambda n: ((n - 1) * 20 + 19) / n        # footnote 6, p. 75
assert 19.5 <= avg(2) and avg(10**6) < 20 and abs(0.75 * avg(2) - 14.625) < 1e-12
```

3 · Cournot → L02-03-cournot

```python
def cournot_br(a, b, ci, qj):
    """Eq. (5.2): BR_i(q_j) = max{0, (a - b q_j - c_i)/(2b)}."""
    return max(0.0, (a - ci - b * qj) / (2 * b))


def cournot_nash(a, b, c1, c2):
    """Unique Cournot-Nash equilibrium of the linear duopoly, corners included."""
    q1, q2 = (a - 2 * c1 + c2) / (3 * b), (a - 2 * c2 + c1) / (3 * b)
    if q1 >= 0 and q2 >= 0:
        return q1, q2
    if q1 < 0:
        q2 = max(0.0, (a - c2) / (2 * b))
        return cournot_br(a, b, c1, q2), q2
    q1 = max(0.0, (a - c1) / (2 * b))
    return q1, cournot_br(a, b, c2, q1)


profit = lambda a, b, ci, qi, qj: (a - b * (qi + qj) - ci) * qi

q1, q2 = cournot_nash(100, 1, 10, 10)
assert (q1, q2) == (30, 30) and 100 - 60 == 40 and profit(100, 1, 10, 30, 30) == 900
 ↪  # pp. 87, 90
assert cournot_br(100, 1, 10, q2) == q1 and cournot_br(100, 1, 10, q1) == q2
 ↪  # mutual BR
# monopoly 45 at 55, profit 2025 (p. 89); collusion at 22.5 each is not a best reply
 ↪  (Gibbons p. 17)
assert profit(100, 1, 10, 45, 0) == 2025 and cournot_br(100, 1, 10, 22.5) == 33.75
assert profit(100, 1, 10, 33.75, 22.5) > profit(100, 1, 10, 22.5, 22.5)
# asymmetric costs c1 = 1, c2 = 2: pi_i = b q_i^2; the gap q1 - q2 = (c2 - c1)/b
q1, q2 = cournot_nash(100, 1, 1, 2)
assert abs(q1 - 100 / 3) < 1e-12 and abs(q1 - q2 - 1) < 1e-12
assert abs(profit(100, 1, 1, q1, q2) - q1 ** 2) < 1e-9
# corner: c1 >= (a + c2)/2 -> firm 1 out, firm 2 a monopolist
assert cournot_nash(100, 1, 55, 10) == (0.0, 45.0) and cournot_nash(100, 1, 70, 10)
 ↪  == (0.0, 45.0)
# n firms: q = (a - c)/((n+1) b), price -> c (Tadelis ex. 5.8 / Gibbons 1.4 setting)
n_firm_price = lambda n, a=100, c=10: (a + n * c) / (n + 1)
assert n_firm_price(1) == 55 and n_firm_price(2) == 40 and n_firm_price(10**6) - 10 <
 ↪  1e-3
```

---

4 · Bertrand on a cent grid → L02-04-bertrand

```python
def bertrand_profit(pi, pj, c=10.0, a=100.0):
    """v_i of p. 88 (homogeneous goods, ties split); vectorised in p_i."""
    pi = np.asarray(pi, float)
```

```
    full = (a - pi) * (pi - c)
    return np.where(pi < pj - 1e-9, full, np.where(np.abs(pi - pj) <= 1e-9, full / 2,
     ↪  0.0))


grid = np.round(np.arange(0, 6001) / 100, 2)          # 0.00 .. 60.00 in cents
BR = []
for pj in grid:
    v = bertrand_profit(grid, pj)
    BR.append(set(np.round(grid[np.abs(v - v.max()) <= 1e-9], 2).tolist()))
idx = {p: i for i, p in enumerate(grid.tolist())}
assert BR[idx[57.0]] == {55.0} and BR[idx[30.0]] == {29.99} and BR[idx[10.01]] ==
 ↪  {10.01}  # p. 90
assert min(BR[idx[10.0]]) == 10.0 and len(BR[idx[10.0]]) == 5001
 ↪  # any p >= 10
NE = sorted((p1, p2) for p1 in grid.tolist() for p2 in BR[idx[p1]] if p1 in
 ↪  BR[idx[p2]])
assert NE == [(10.0, 10.0), (10.01, 10.01)]
 ↪  # p. 90
# Delta v of p. 89 at p_j = 10.02: 0.0002 with the book's rounding, 0.0001 exactly
dv_book = 55.02 * 10.02 - 0.5 * 10.02 ** 2 - 501.1
dv_exact = 89.99 * 0.01 - 0.5 * 89.98 * 0.02
assert abs(dv_book - 0.0002) < 1e-9 and abs(dv_exact - 0.0001) < 1e-9
# asymmetric, continuous: v1(p1, 2) -> 98 as p1 -> 2 from below, 49 at 2, 0 above
 ↪  (pp. 91-92)
v1 = lambda p1: (100 - p1) * (p1 - 1) if p1 < 2 else (49.0 if p1 == 2 else 0.0)
assert v1(2 - 1e-9) < 98 and 98 - v1(2 - 1e-9) < 1e-6 and v1(2) == 49 and v1(2.5) ==
 ↪  0
# differentiated Bertrand (Gibbons p. 22): p* = (a + c)/(2 - b) solves p = (a + b p +
 ↪  c)/2
a_, b_, c_ = 10.0, 0.5, 2.0
p_star = (a_ + c_) / (2 - b_)
assert abs(p_star - (a_ + b_ * p_star + c_) / 2) < 1e-12
```

---

5 · The commons → L02-05-commons

```
import math

K = 1.0
k_ne = K / 3                          # k_i = (K - k_j)/2 at both (p. 86)
assert abs(k_ne - (K - k_ne) / 2) < 1e-15
v_sym = lambda k: math.log(k) + math.log(K - 2 * k)
# the symmetric social optimum K/4 (p. 87): grid maximiser of v(k, k)
ks = np.linspace(0.01, 0.49, 48001)
assert abs(ks[np.argmax([v_sym(k) for k in ks])] - 0.25) < 1e-4
gain = v_sym(K / 4) - v_sym(K / 3)
assert abs(gain - math.log(9 / 8)) < 1e-12 and round(gain, 3) == 0.118
# footnote 5 prints 0.051: that is the base-10 value of the same expression
assert round(2 * math.log10(3) - math.log10(2) - math.log10(4), 3) == 0.051
```

6 · Hotelling voting → L02-06-hotelling-voting

```
PLAT = range(-50, 51)


def vote_payoff(ai, aj):
    """1 win, 1/2 tie, 0 lose; equidistant citizens abstain (Tadelis fn. 9, p.
    ↪ 94)."""
    if ai == aj:
        return 0.5
    mine = sum(abs(x - ai) < abs(x - aj) for x in PLAT)
    theirs = sum(abs(x - aj) < abs(x - ai) for x in PLAT)
    return 1.0 if mine > theirs else 0.5 if mine == theirs else 0.0


assert sum(abs(x + 15) < abs(x - 22) for x in PLAT) == 54          # Fig. 5.4:
↪ citizens -50..3
PAY = {(i, j): vote_payoff(i, j) for i in PLAT for j in PLAT}
BRv = {j: [i for i in PLAT if PAY[i, j] == max(PAY[k, j] for k in PLAT)] for j in
↪ PLAT}
assert BRv[22] == list(range(-21, 22)) and BRv[-15] == list(range(-14, 15))
assert BRv[0] == BRv[1] == BRv[-1] == [0]                            # footnote 10
assert [(i, j) for j in PLAT for i in BRv[j] if j in BRv[i]] == [(0, 0)]
```

---

7 · Mixed equilibria: indifference and support enumeration → L02-07-mixed-strategies

```
def mixed_2x2(U1, U2):
    """Totally mixed NE of a 2x2 game from the two indifference conditions (Prop.
    ↪ 6.1).

    Returns (p, q) = (Pr row 1, Pr column 1) as Fractions, or None if not in (0,1)^2.
    """
    (a, b), (c, d) = [[Fr(x) for x in r] for r in U1]
    (e, f), (g, h) = [[Fr(x) for x in r] for r in U2]
    # player 1 indifferent: q a + (1-q) b = q c + (1-q) d
    den1, den2 = (a - c) - (b - d), (e - f) - (g - h)
    if den1 == 0 or den2 == 0:
        return None
    q = (d - b) / den1
    p = (h - g) / den2                    # player 2 indifferent: p e + (1-p) g = p f
↪ + (1-p) h
    return (p, q) if 0 < p < 1 and 0 < q < 1 else None


assert mixed_2x2([[1, -1], [-1, 1]], [[-1, 1], [1, -1]]) == (Fr(1, 2), Fr(1, 2))
↪ # p. 111
```

```
assert mixed_2x2([[0, 3], [4, 0]], [[0, 5], [4, 3]]) == (Fr(1, 6), Fr(3, 7))
 ↪ # p. 113
assert mixed_2x2([[2, 0], [0, 1]], [[1, 0], [0, 2]]) == (Fr(2, 3), Fr(1, 3))
 ↪ # BoS
assert mixed_2x2([[5, 0], [3, 3]], [[5, 3], [0, 3]]) == (Fr(3, 5), Fr(3, 5))
 ↪ # Stag Hunt


def support_enumeration(U1, U2):
    """All NE of a nondegenerate bimatrix game (MWG Prop. 8.D.1): equal payoffs on
     ↪ the
    support, no better payoff off it. Exact arithmetic; returns list of (sigma1,
 ↪ sigma2)."""
    U1 = [[Fr(x) for x in r] for r in U1]
    U2 = [[Fr(x) for x in r] for r in U2]
    m, n = len(U1), len(U1[0])
    out = []

    def solve(A, rhs):
        """Square system by Gaussian elimination over Fractions; None if singular."""
        A = [row[:] + [v] for row, v in zip(A, rhs)]
        k = len(A)
        for col in range(k):
            piv = next((r for r in range(col, k) if A[r][col] != 0), None)
            if piv is None:
                return None
            A[col], A[piv] = A[piv], A[col]
            A[col] = [x / A[col][col] for x in A[col]]
            for r in range(k):
                if r != col and A[r][col] != 0:
                    A[r] = [x - A[r][col] * y for x, y in zip(A[r], A[col])]
        return [A[i][-1] for i in range(k)]

    for k in range(1, min(m, n) + 1):
        for S1 in itertools.combinations(range(m), k):
            for S2 in itertools.combinations(range(n), k):
                # sigma2 on S2 makes player 1 indifferent over S1 (unknowns: probs +
                 ↪ value)
                A = [[U1[r][c] for c in S2] + [Fr(-1)] for r in S1] + [[Fr(1)] * k +
 ↪ [Fr(0)]]
                x2 = solve(A, [Fr(0)] * k + [Fr(1)])
                A = [[U2[r][c] for r in S1] + [Fr(-1)] for c in S2] + [[Fr(1)] * k +
 ↪ [Fr(0)]]
                x1 = solve(A, [Fr(0)] * k + [Fr(1)])
                if x1 is None or x2 is None or min(x1[:-1]) < 0 or min(x2[:-1]) < 0:
                    continue
                s1 = [Fr(0)] * m
                s2 = [Fr(0)] * n
                for i, r in enumerate(S1):
                    s1[r] = x1[i]
```

```
                for i, c in enumerate(S2):
                    s2[c] = x2[i]
                v1 = [sum(U1[r][c] * s2[c] for c in range(n)) for r in range(m)]
                v2 = [sum(U2[r][c] * s1[r] for r in range(m)) for c in range(n)]
                if max(v1) == x2[-1] and max(v2) == x1[-1]:        # nothing
                 ↪ off-support does better
                    out.append((tuple(s1), tuple(s2)))
    return out


RPS1 = [[0, -1, 1], [1, 0, -1], [-1, 1, 0]]
RPS2 = [[-x for x in r] for r in RPS1]
third = (Fr(1, 3),) * 3
assert support_enumeration(RPS1, RPS2) == [(third, third)]                          #
 ↪ pp. 111-113
# p. 81: (M, C) is the unique PURE NE, but the game has five NE in all (four mixed)
eq81 = support_enumeration(U1, U2)
s16 = (Fr(1, 6), Fr(5, 6), Fr(0))
assert len(eq81) == 5 and ((0, 1, 0), (0, 1, 0)) in eq81 and (s16, s16) in eq81
# a degenerate game (1 for both at (first, first), else 0): two pure NE and no mixed
 ↪ one,
# the exception to "two pure => a third, mixed" (pitfalls item 17; the companion's
 ↪ exact solver agrees)
weak = support_enumeration([[1, 0], [0, 0]], [[1, 0], [0, 0]])
assert set(weak) == {((1, 0), (1, 0)), ((0, 1), (0, 1))}
eq = support_enumeration([[0, 3], [4, 0]], [[0, 5], [4, 3]])
assert len(eq) == 3 and ((Fr(1, 6), Fr(5, 6)), (Fr(3, 7), Fr(4, 7))) in eq
 ↪ # p. 113
# RPS payoffs against (1/2, 1/2, 0): R -1/2, P 1/2, S 0 (p. 106)
s2 = [Fr(1, 2), Fr(1, 2), Fr(0)]
assert [sum(RPS1[r][c] * s2[c] for c in range(3)) for r in range(3)] == [Fr(-1, 2),
 ↪ Fr(1, 2), 0]
# all-pay auction against U[0,1] (p. 107): every bid in [0,1] earns 0
assert all(abs((1 - s) * (-s) + s * (1 - s)) < 1e-15 for s in np.linspace(0, 1, 101))
```

---

8 · Dominance by a mixed strategy (linear program) → L02-08-iesds-rationalizability-revisited

```
from scipy.optimize import linprog


def dominated_by_mix(U, s, others):
    """Is pure row `s` strictly dominated by a mix over `others` (Def. 6.7)?

    U[r][c]: the player's payoff, own strategies as rows. Maximise eps subject to
    sum_r x_r U[r][c] >= U[s][c] + eps for every column c, x in the simplex.
    Returns (eps, x); dominated iff eps > 0.
    """
    U = np.asarray(U, float)
```

```
    k, n = len(others), U.shape[1]
    c = np.zeros(k + 1)
    c[-1] = -1.0                                          # maximise eps
    A_ub = np.hstack([-U[others].T, np.ones((n, 1))])
    b_ub = -U[s]
    A_eq = np.hstack([np.ones((1, k)), np.zeros((1, 1))])
    res = linprog(c, A_ub=A_ub, b_ub=b_ub, A_eq=A_eq, b_eq=[1.0],
                  bounds=[(0, 1)] * k + [(None, None)], method="highs")
    return res.x[-1], res.x[:-1]


# Tadelis p. 115: player 2's L is dominated by a C/R mix; (0, 1/2, 1/2) gives 2, 2.5,
 ↪  3.5
V2 = np.array([[1, 4, 0], [2, 0, 5], [3, 4, 3]], float).T           # player 2's
 ↪  payoffs, own strategies as rows
eps, x = dominated_by_mix(V2, 0, [1, 2])
assert eps > 0 and np.allclose(0.5 * V2[1] + 0.5 * V2[2], [2, 2.5, 3.5])
# no pure strategy dominates L (pure IESDS does nothing)
assert not any(all(V2[j] > V2[0]) for j in (1, 2))
# p. 116: after L is gone, U is dominated by (0, 1/2, 1/2) over M, D: 2 and 1.5
 ↪  against 1 and 1
V1 = np.array([[1, 1], [0, 3], [4, 0]], float)                       # player 1,
 ↪  columns C, R
eps, _ = dominated_by_mix(V1, 0, [1, 2])
assert eps > 0 and np.allclose(0.5 * V1[1] + 0.5 * V1[2], [2, 1.5])
# ...but not before L is removed (against L, U's 5 beats the mix's 3.5)
V1full = np.array([[5, 1, 1], [3, 0, 3], [4, 4, 0]], float)
assert dominated_by_mix(V1full, 0, [1, 2])[0] <= 1e-9
# Gibbons p. 32: B = (1, 1) dominated by T/2 + M/2; B = (2, 2) is not dominated
assert dominated_by_mix([[3, 0], [0, 3], [1, 1]], 2, [0, 1])[0] > 0
assert dominated_by_mix([[3, 0], [0, 3], [2, 2]], 2, [0, 1])[0] <= 1e-9
```

---

9 · Equilibria as fixed points of BR → L02-09-existence

```
def br1(q):  # Tadelis (6.3), p. 119: p as a set, the 6.2.3 game
    return {1.0} if q < 3 / 7 else ({0.0} if q > 3 / 7 else "all")


def br2(p):  # (6.4)
    return {1.0} if p < 1 / 6 else ({0.0} if p > 1 / 6 else "all")


def is_fixed(q, p):
    b1, b2 = br1(q), br2(p)
    return (b1 == "all" or p in b1) and (b2 == "all" or q in b2)


pts = [(q, p) for q in np.linspace(0, 1, 701) for p in np.linspace(0, 1, 601) if
 ↪  is_fixed(q, p)]
```

```
assert sorted(pts) == [(0.0, 1.0), (1.0, 0.0)]          # the grid misses 3/7 and 1/6
 ↪  ...
assert is_fixed(3 / 7, 1 / 6)                            # ... the mixed fixed point
 ↪  is exact
assert br1(0.8) == {0.0} and br2(0.5) == {0.0}           # (q1, p1) -> (0, 0), Fig.
 ↪  6.6
print("Lecture 2 python note: all checks pass")
```

The last block also shows why the companion and these checks solve for equilibria **exactly** rather than on a grid: the mixed fixed point $(\frac{3}{7}, \frac{1}{6})$ is not on any decimal grid, and a grid search silently loses it.

Links

- Figures: `figures/make_figures.py` (run: `python Notas/L02/figures/make_figures.py`).
- Companion checks: `companions/check_companions.py` (run: `python Notas/L02/companions/check_compan`
- Lecture 1 code: `Listas/lista-lecture-01-check.py`.